

IT Consultancy

Lecture 4

Framing the Problem

AS-IS Modelling

Designing the Analyses

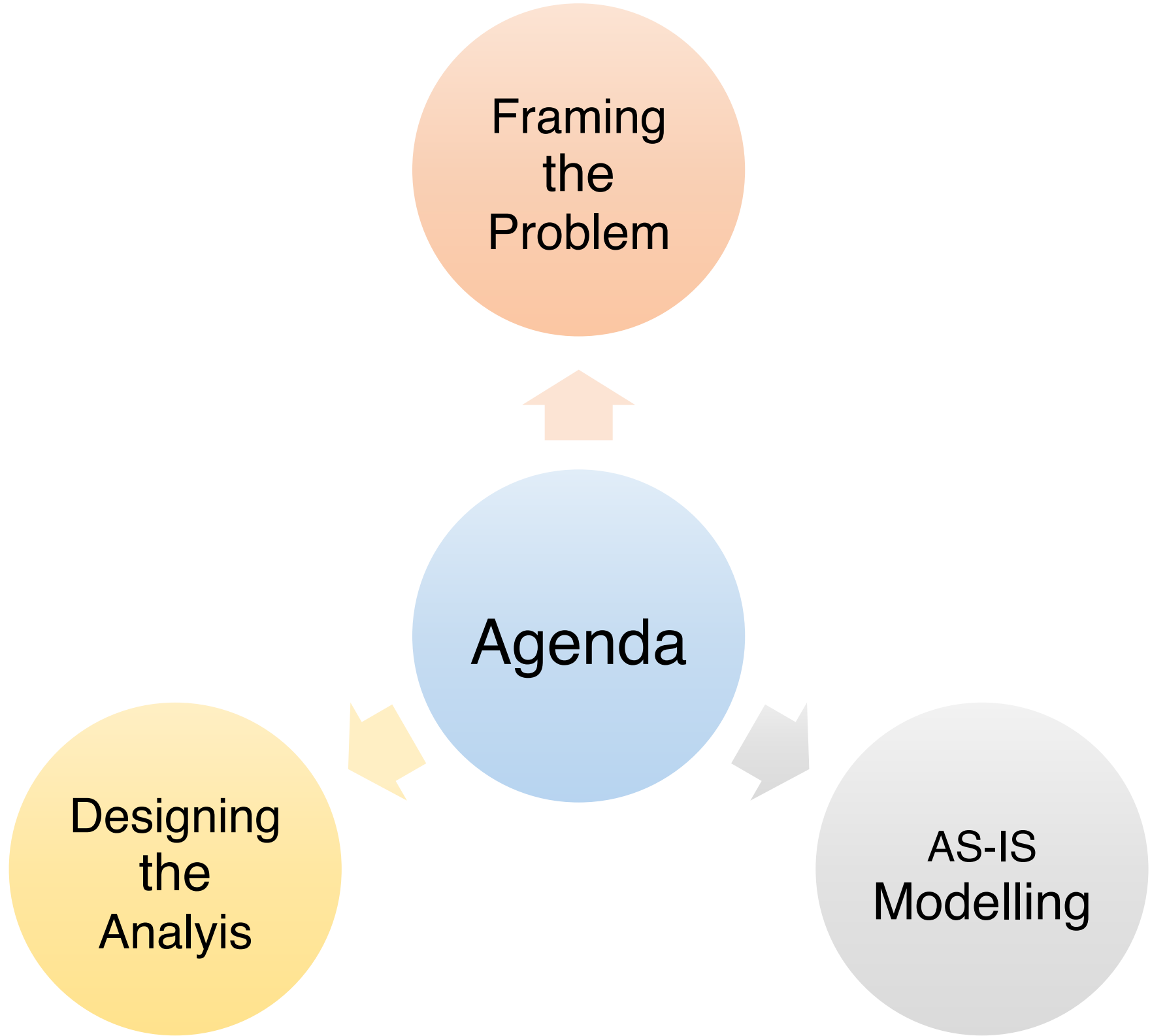


Enterprise Designers by Jay Forrester

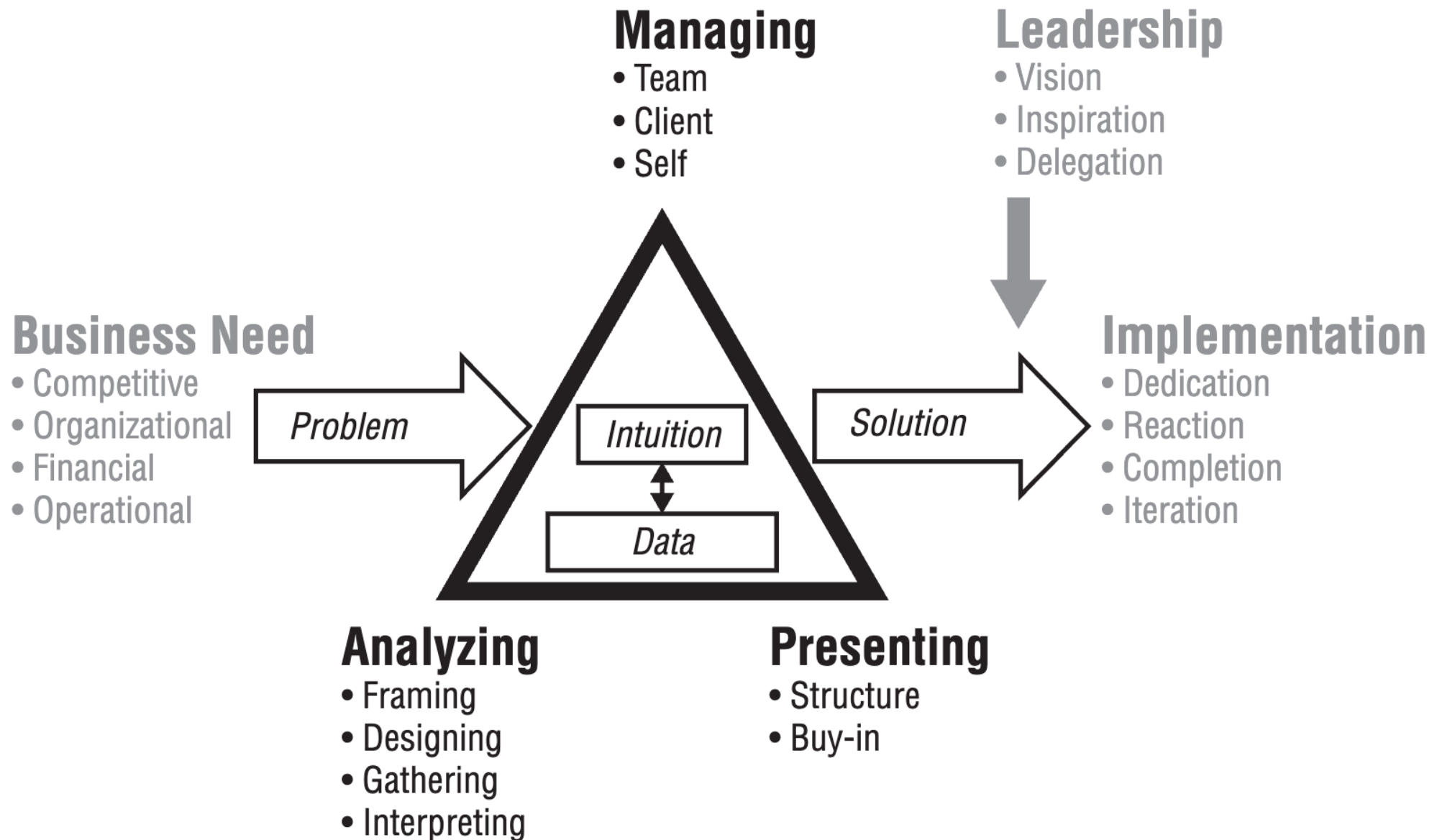
“A fundamental difference exists between an enterprise operator and an enterprise designer. To illustrate, consider the two most important people in successful operation of an airplane. One is the airplane designer and the other is the airplane pilot. The designer creates an airplane that ordinary pilots can fly successfully. Is not the usual manager more a pilot than a designer? A manager runs an organization, just as a pilot runs an airplane. Success of a pilot depends on an aircraft designer who created a successful airplane. On the other hand, who designed the corporation that a manager runs? Almost never has anyone intentionally and thoughtfully designed an organization to achieve planned growth and stability.”

Enterprise Designers by Jay Forrester

“Education, in present management schools, trains operators of corporations. There is almost no attention to designing corporations. Corporate successes and failures seldom arise from functional specialties alone. Corporate performance grows out of the interactions among functional specialties. Present day management education fails to convey the importance of how parts of a business interact with one another and with the outside world. In the future, we must deal with the way policies determine the future of an organization.”



McKinsey Problem Solving Model



Framing the Problem

Structure and MECE

- Frame problems so that you can conduct rigorous fact-based analysis.
- Use structured frameworks to generate fact-based hypotheses.
- What is structure?
 - Problem solving frameworks
 - Defining the problem and breaking it down into component elements
 - Don't reinvent the wheel BUT each client is unique
 - Without structure your ideas won't stand up
 - Use structure to strengthen your thinking
- MECE – *mee-see* – Mutually Exclusive, Collectively Exhaustive

MECE

- Structure issues or ideas clearly without overlap or gaps.
- Break down a problem or topic into non-overlapping categories (mutually exclusive) that together cover the entire scope (collectively exhaustive).
- Example: To improve profitability, break it down into: (a) Increase revenue (b) Reduce costs
- Nothing overlaps; all possibilities are covered

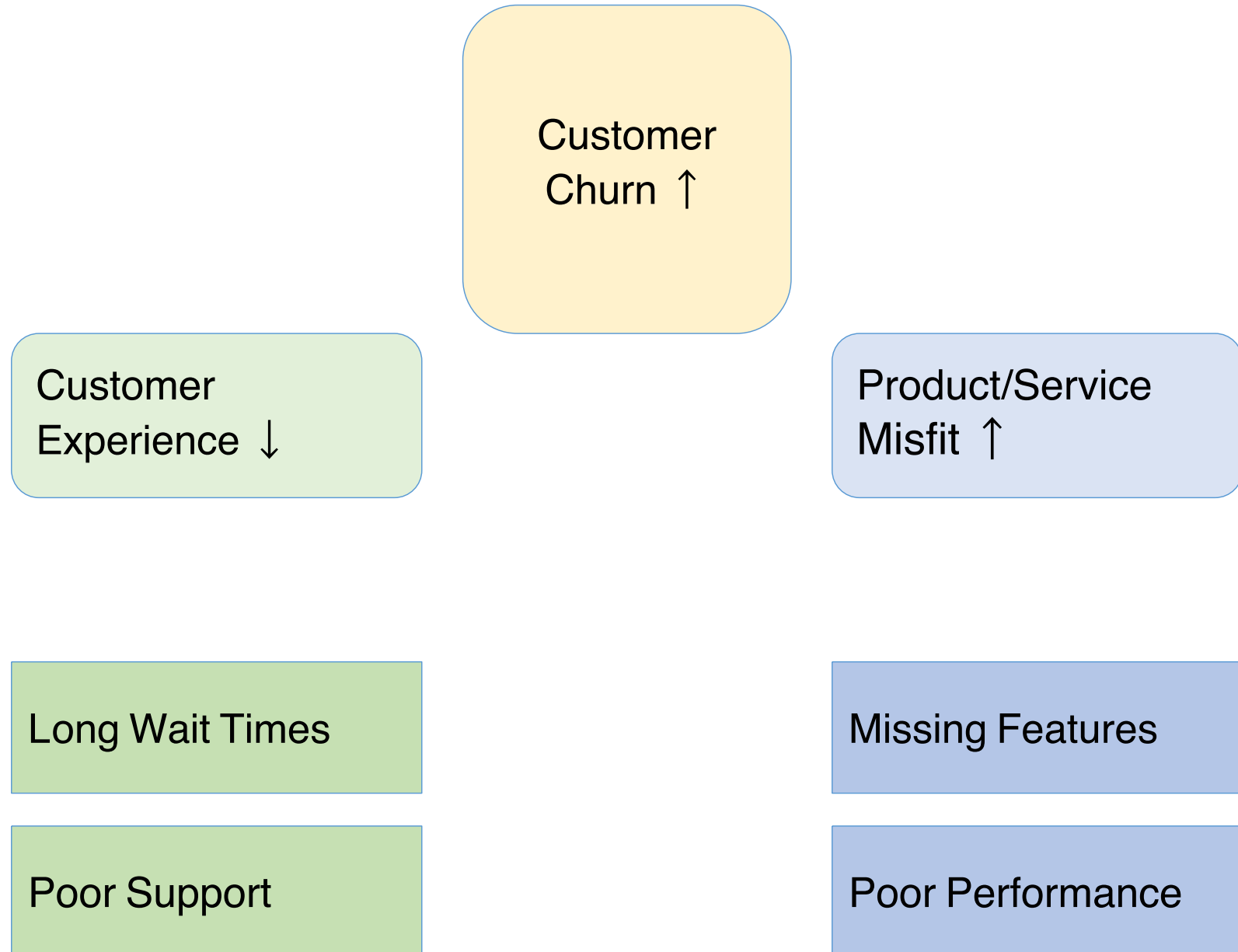
Issue/Logic/Hypotheses Trees

- Decompose a high-level problem into smaller, manageable sub-problems.
- A tree structure where the branches are MECE. Diagnostic and Solution Trees.
- Example - Problem: “Why is profitability declining?”
 - Revenue ↓
 - Volume ↓
 - Price ↓
 - Costs ↑
 - Fixed costs ↑
 - Variable costs ↑
- Used for Root cause analysis, Hypothesis

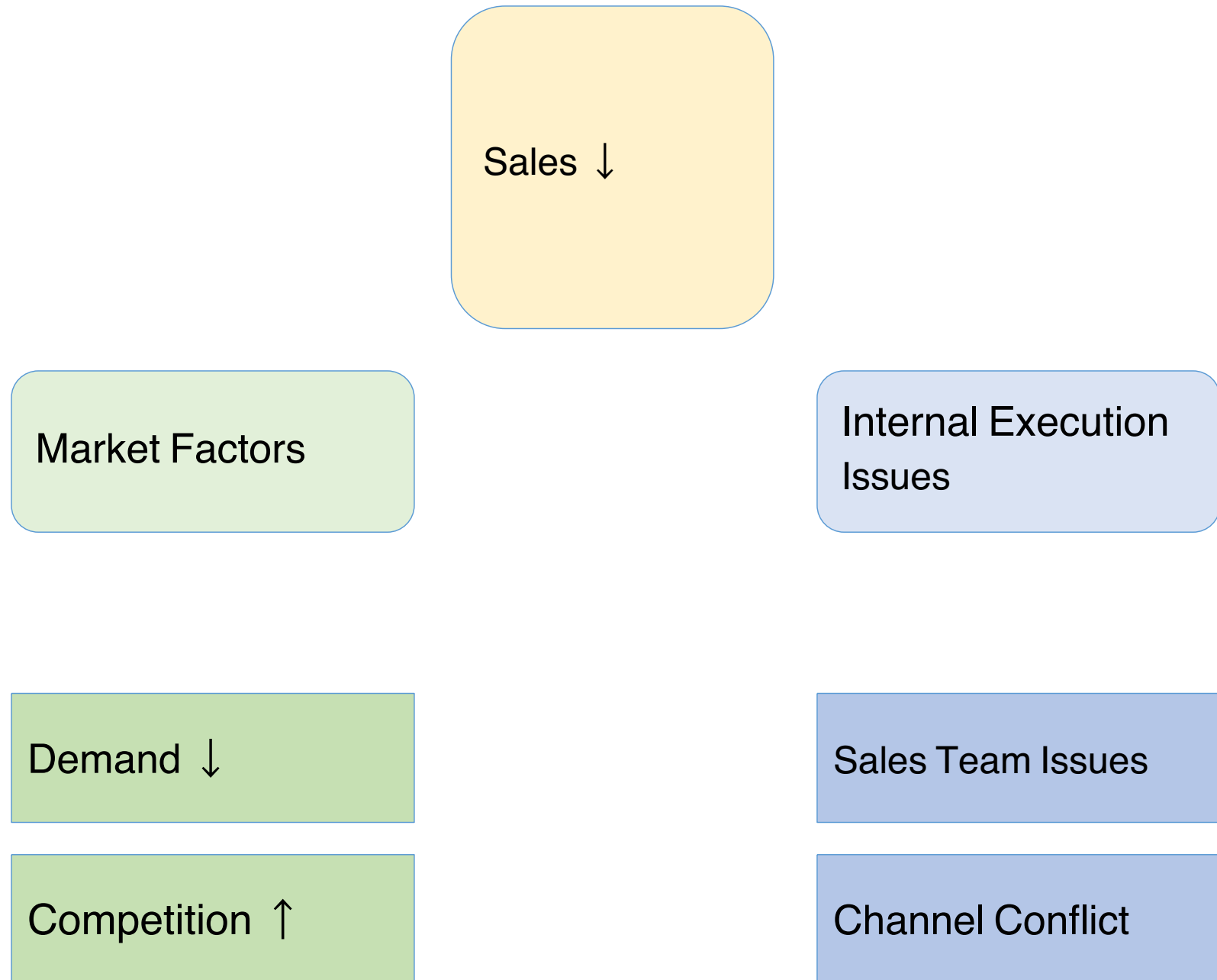
Hypothesis-Driven Approach

- Start with a hypothesis and test it with data
- Focuses effort on proving or disproving what matters most.
- Example Hypothesis: “Customer churn is primarily driven by slow service resolution time.”
- Helps prioritizing data analysis, time-boxing workstreams, avoiding “boiling the ocean”

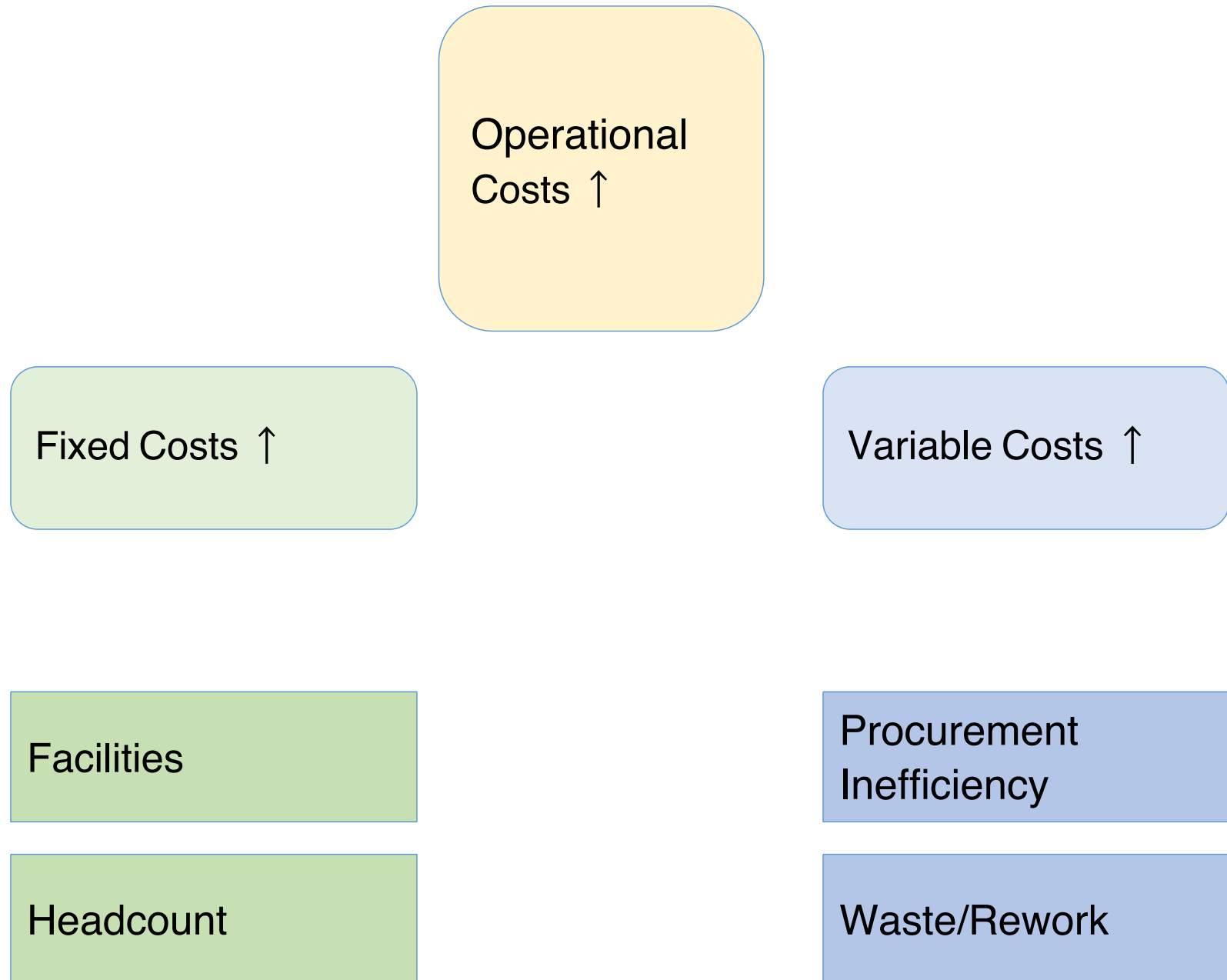
Why is Customer Churn Increasing?



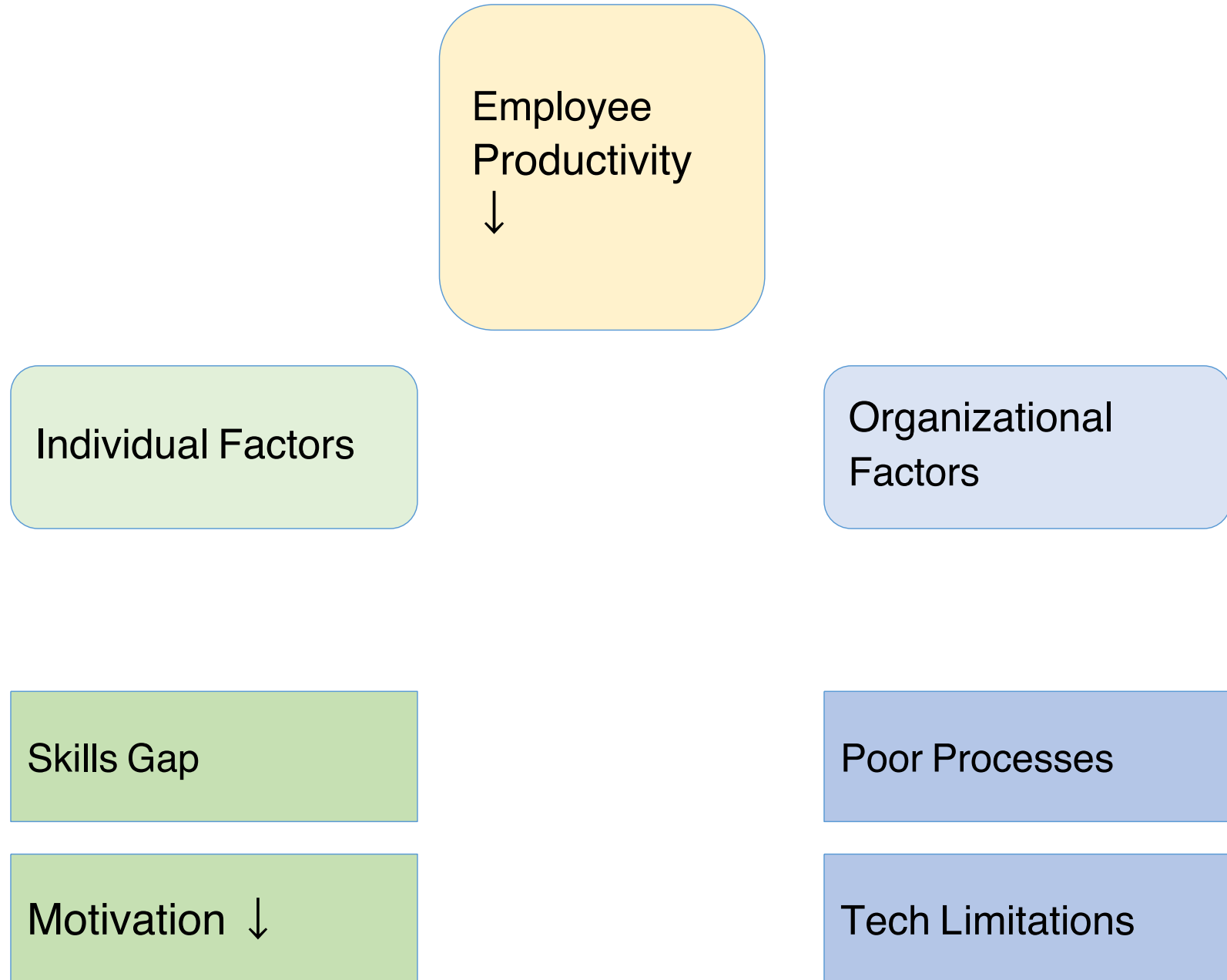
Why are Sales Declining?



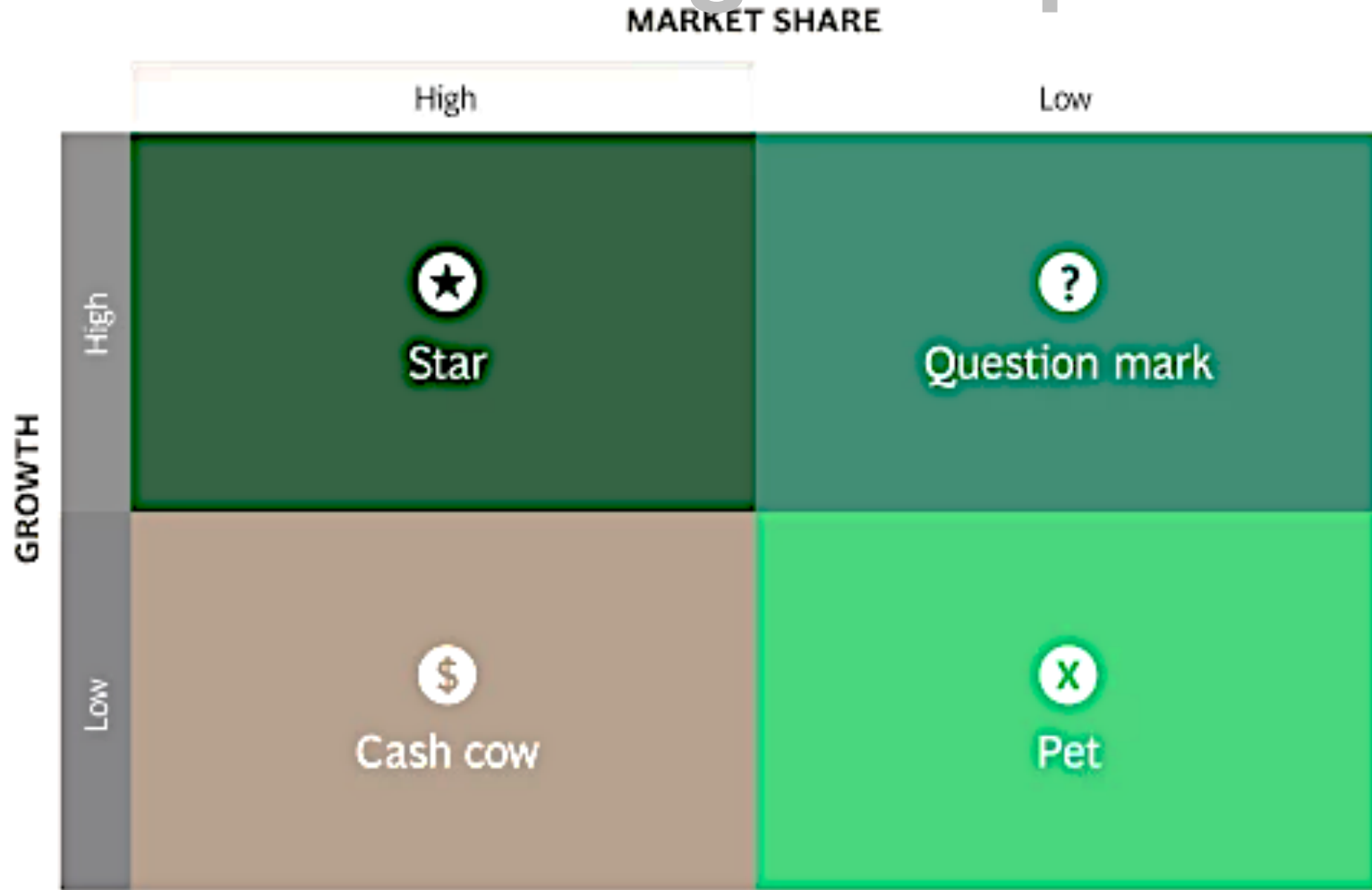
Why are Operational Costs Increasing?



Why is employee productivity low?



Boston Consulting Group Matrix



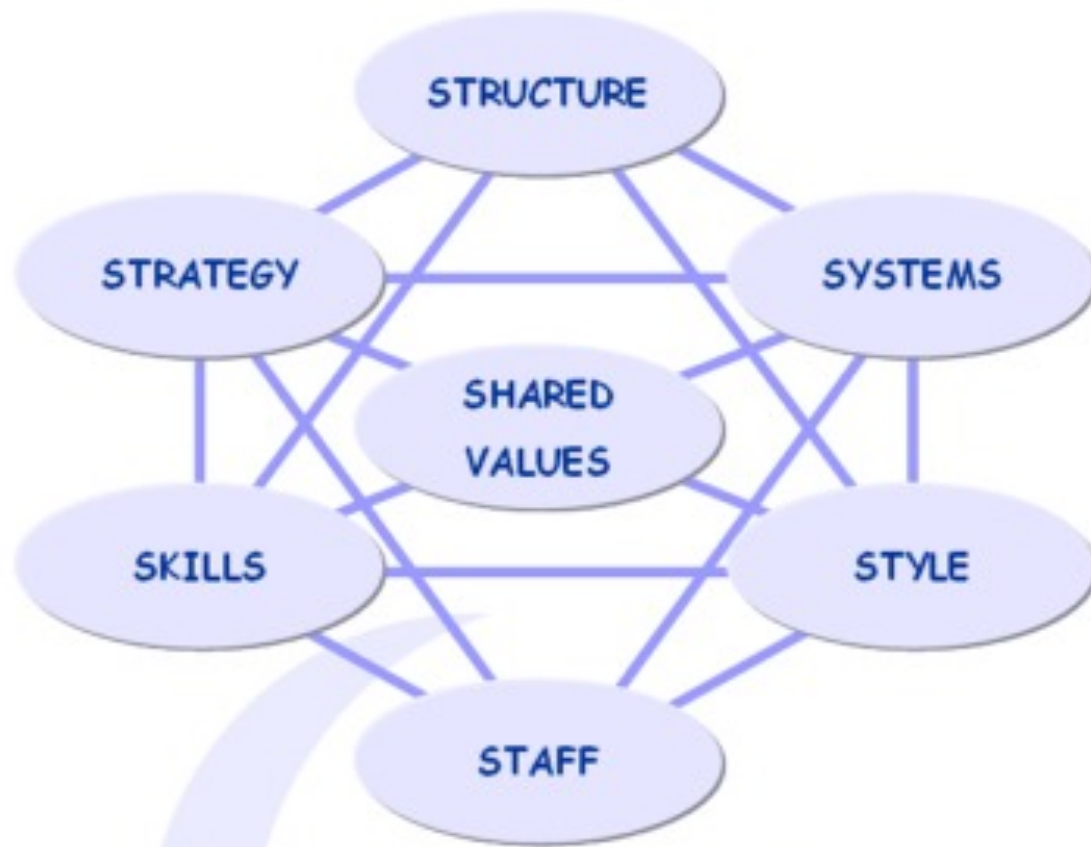
“By assigning each business to one of these four categories, executives could then decide where to focus their resources and capital to generate the most value, as well as where to cut their losses.” *BCG*

Portfolio management

Investment decisions

Reference Models

		Reference models						
		INK	EFQM	ISO 9000:2008	McKinsey 7S	KPMG	Twijnstra EHS	Balanced Scorecard
Building blocks	Strategy Value & Development	Policy and Strategy orientate create implement	Policy and Strategy	Management responsibility	Strategy	Management & Organisation	Strategy	Learning and Growth Perspective
	Structure Processes & Organisation	Processes design control improve	Processes	Quality Management System	Structure	Processes	Structure	Internal Business Perspective
	Culture Leadership & Core values	Leadership	Leadership	Management responsibility	Style, Shared Values	Culture	Mgmt. Style, Culture	Internal Business Perspective
	People Competence development	People organise, invest, respect	People	Resource management	Staff, Skills	People	Personel	Internal Business Perspective
	Resources ICT, Money Facilities	Resources money, knowledge & technology, material & services	Partnerships & Resources	Resource management	Systems	Resources & Systems	Systems	Internal Business Perspective
	Results Products & Effects	Resultaten Waardering klanten, leveranciers, medewerkers, maatschappij Eindresultaten	Key Performance Results	Monitoring Analysis Customer satisfaction	-	Performance	-	Financial Perspective Customer Perspective



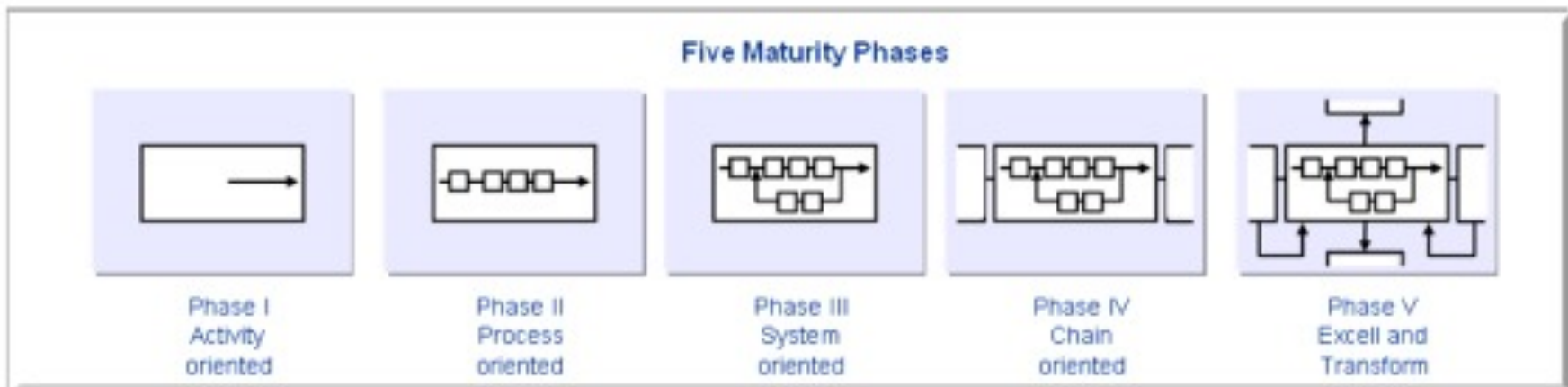
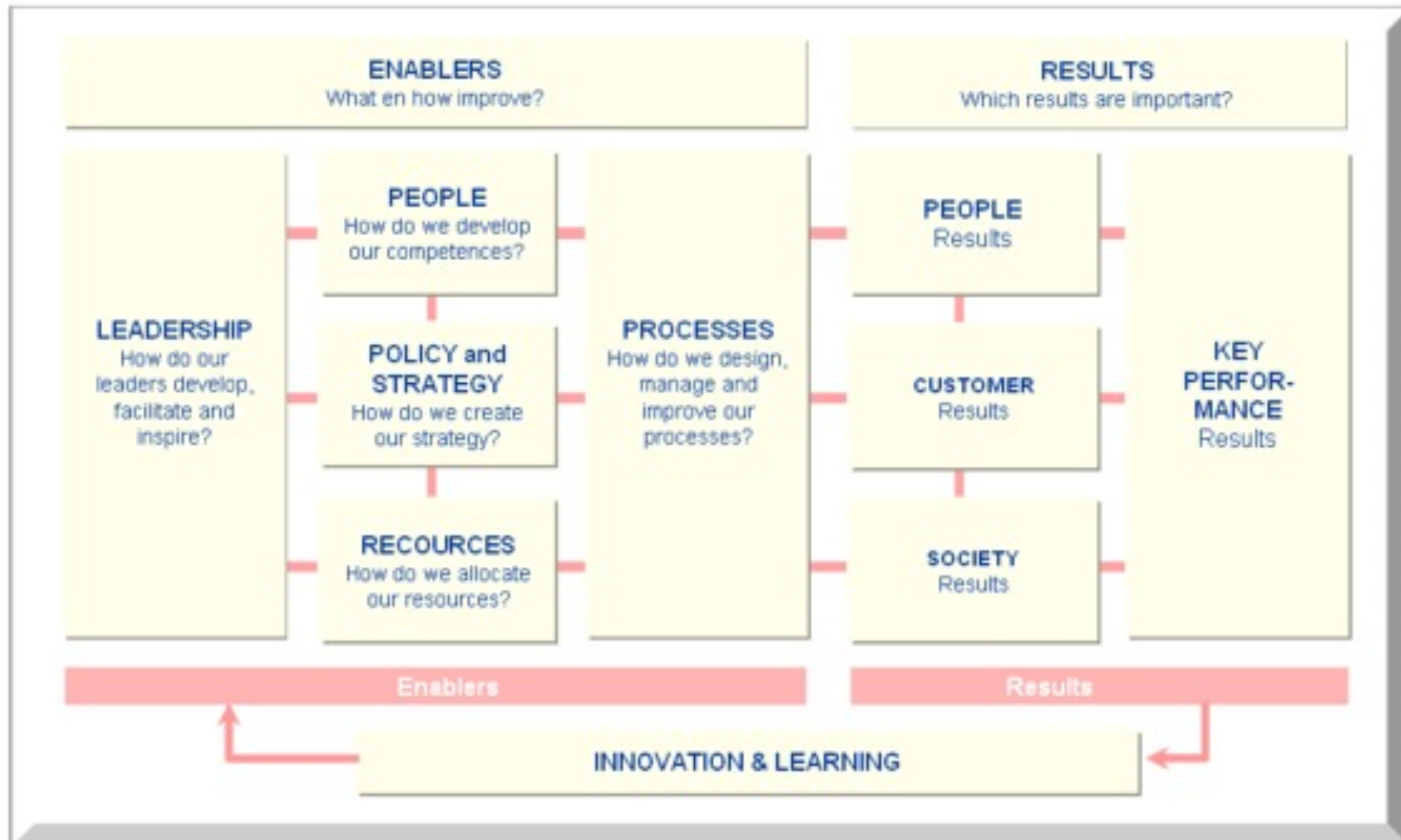
McKinsey

7-S Model

A.G. Athos en R.T. Pascale
The art of Japanese management

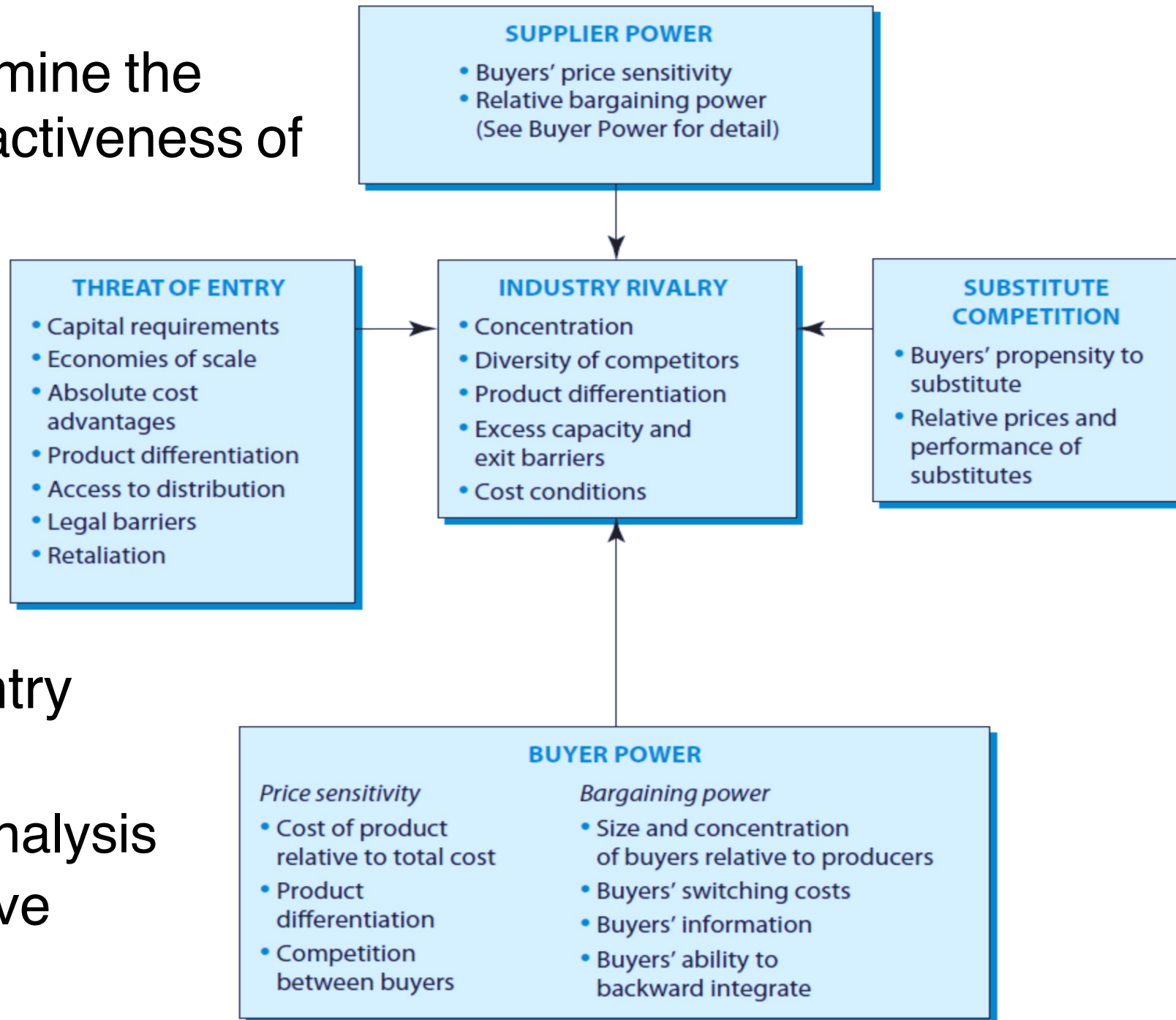
KPMG-Model





Porter's Five Forces Model

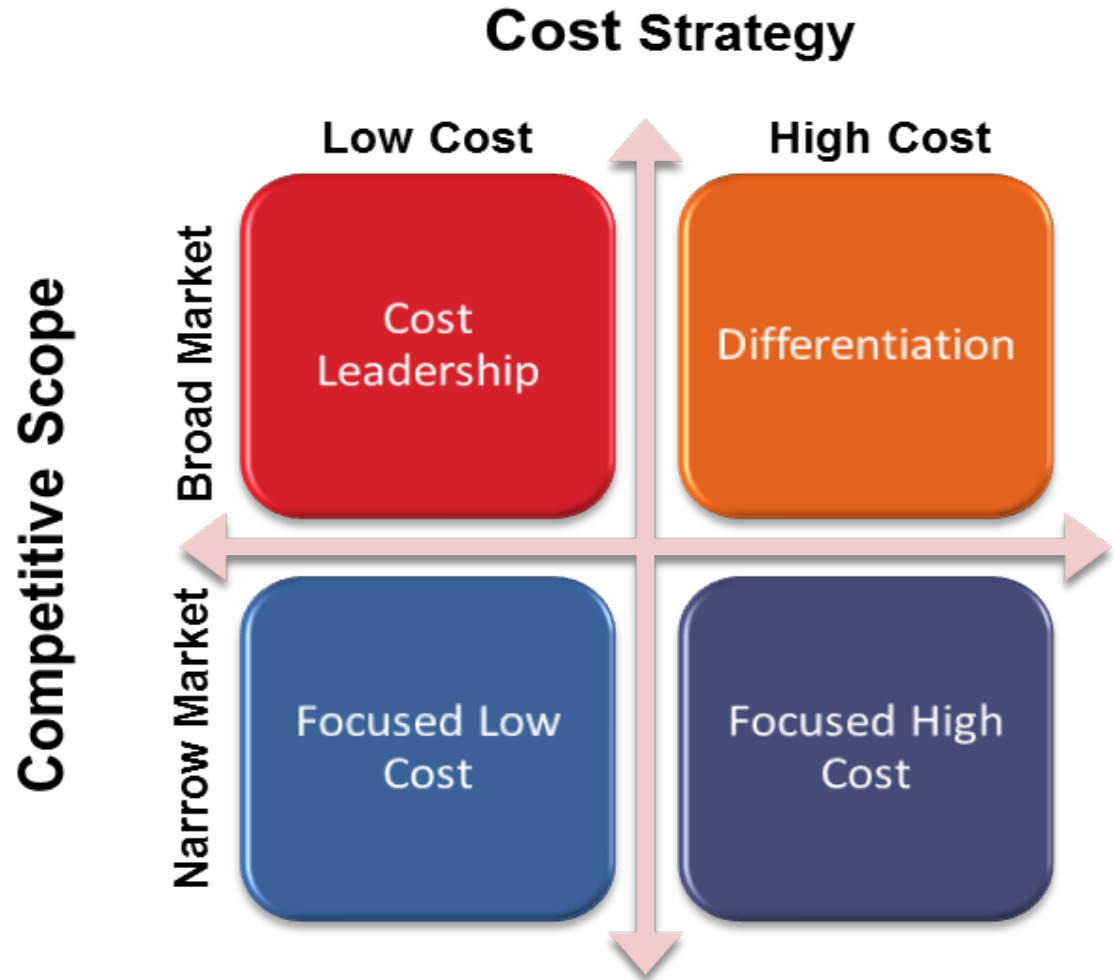
Helps determine the relative attractiveness of an industry.



- Market entry strategy
- Industry analysis
- Competitive strategy

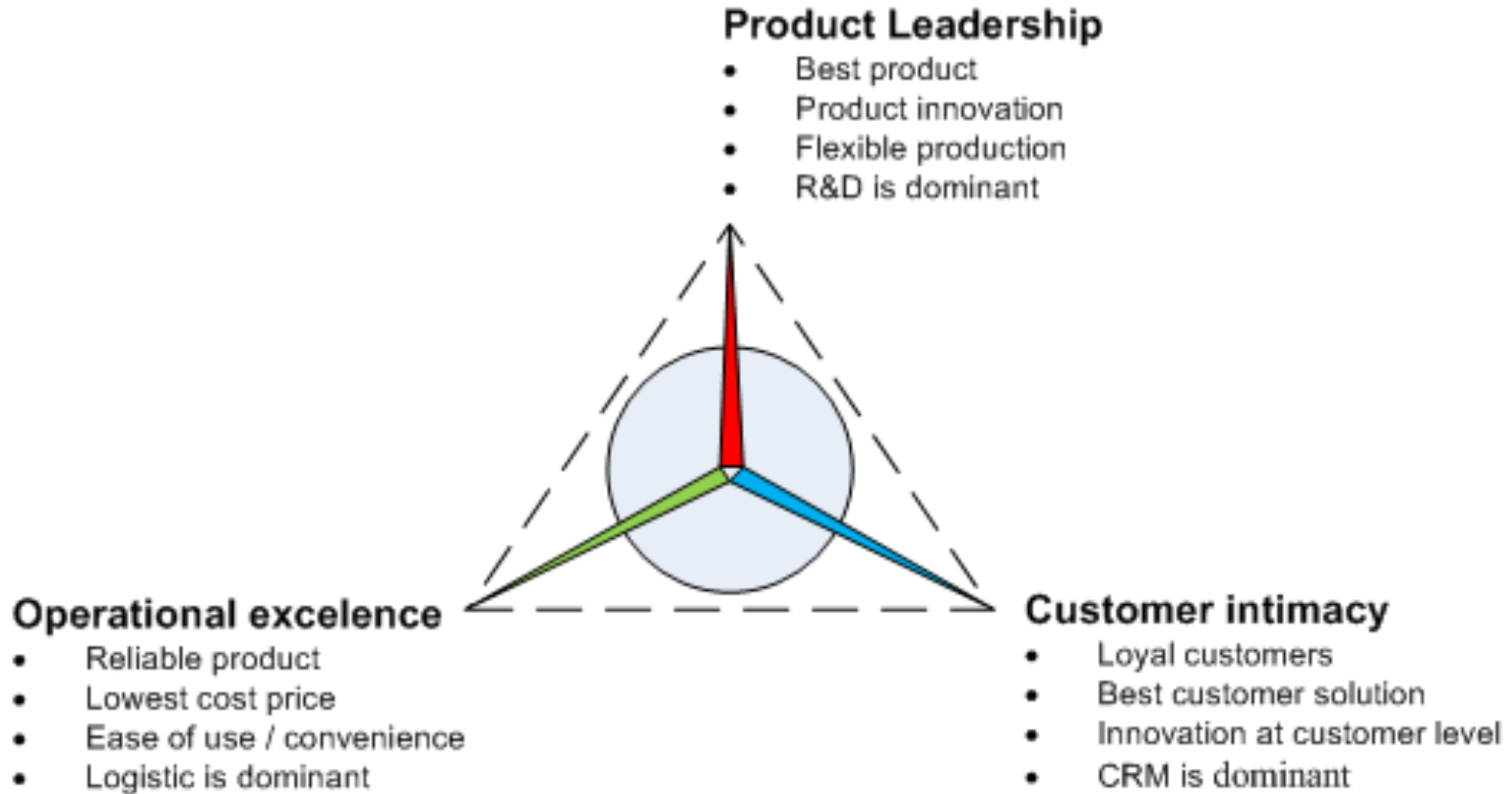
Porter's Generic Strategy Model

Determine a
suitable
Generic
Strategy!



<https://www.youtube.com/watch?v=V14kuqYEsxE>

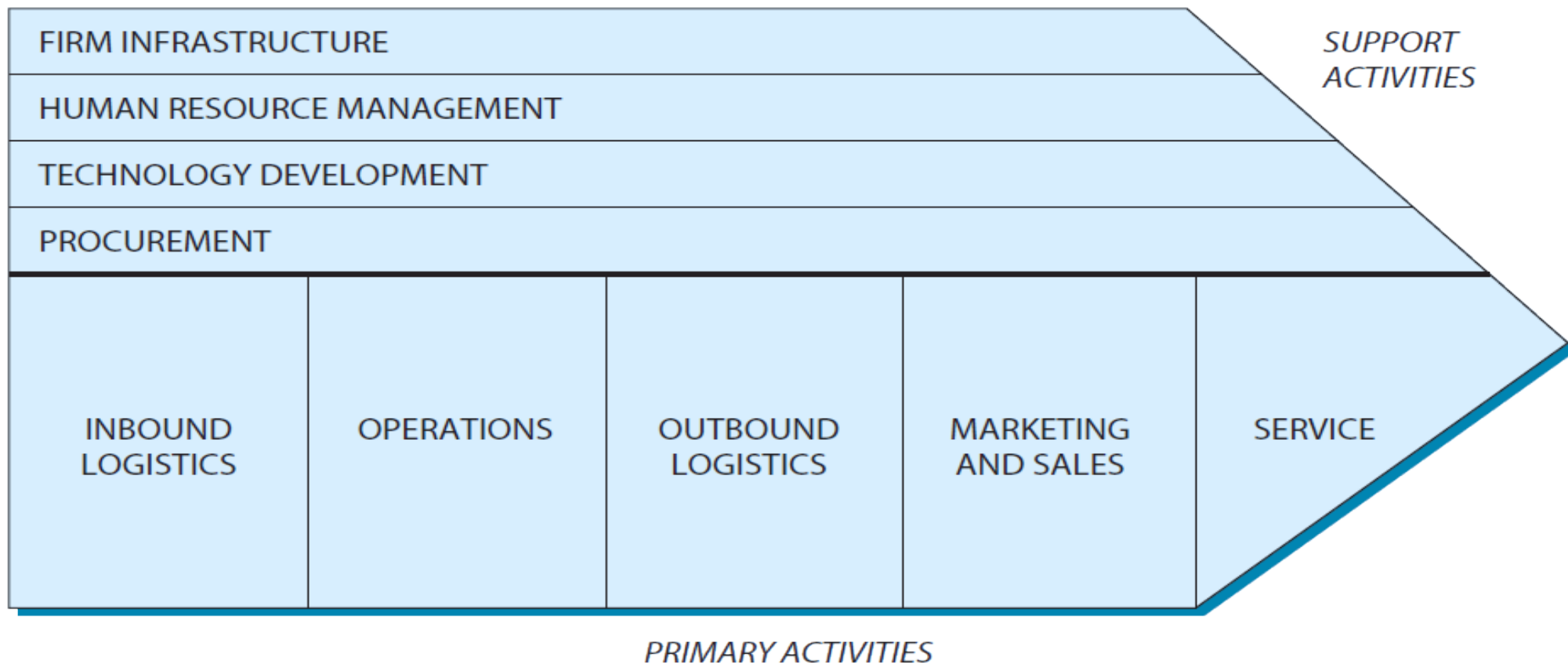
Treacy and Wiersema's Value Disciplines



Value Chain

- Interlinked value-adding activities that convert inputs into outputs which help create competitive advantage(s).

www.businessdictionary.com



Cost reduction

Efficiency improvement

Capability mapping

SWOT Analysis

S TRENGTHS



- Fast return on investment
- Will shorten our time to market for new products 10%

W EAKNESSES



- Not as good ROI as project B
- Don't have enough resources to do the work

Internal factors

O PPORTUNITIES



- Make us more competitive against our main rival
- Can move into new geographic markets

T HREATS



- Have to meet new governance rules in this market
- We think our main competitor is working on a similar project

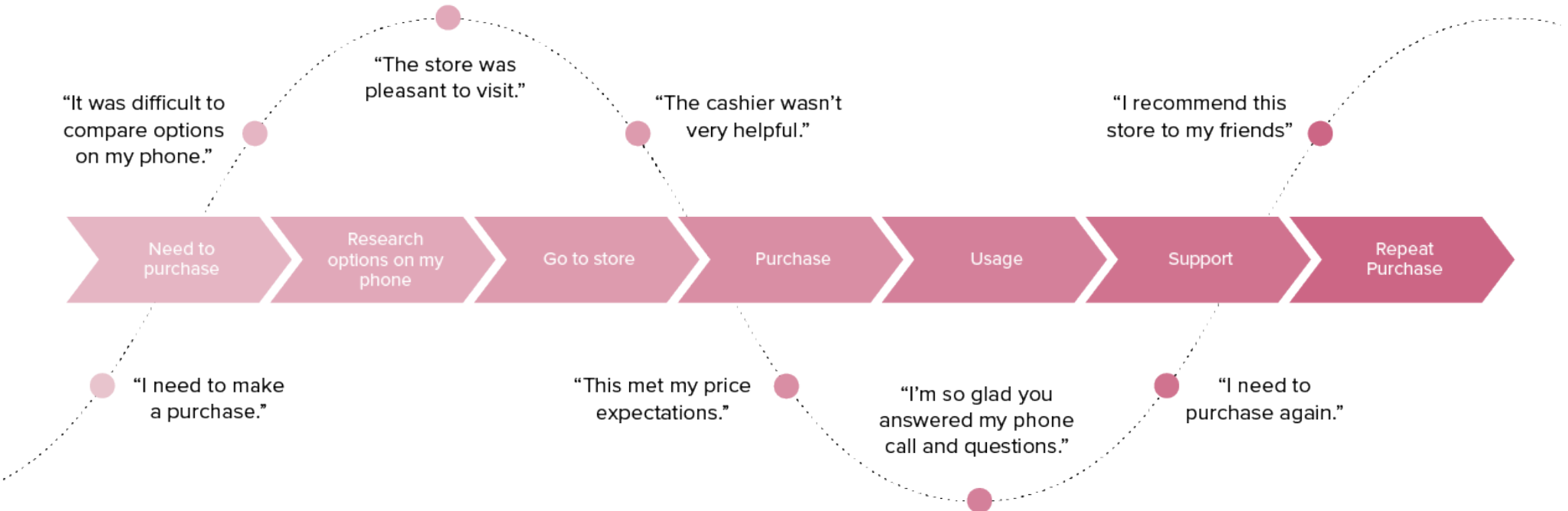
External factors

Positive

Negative

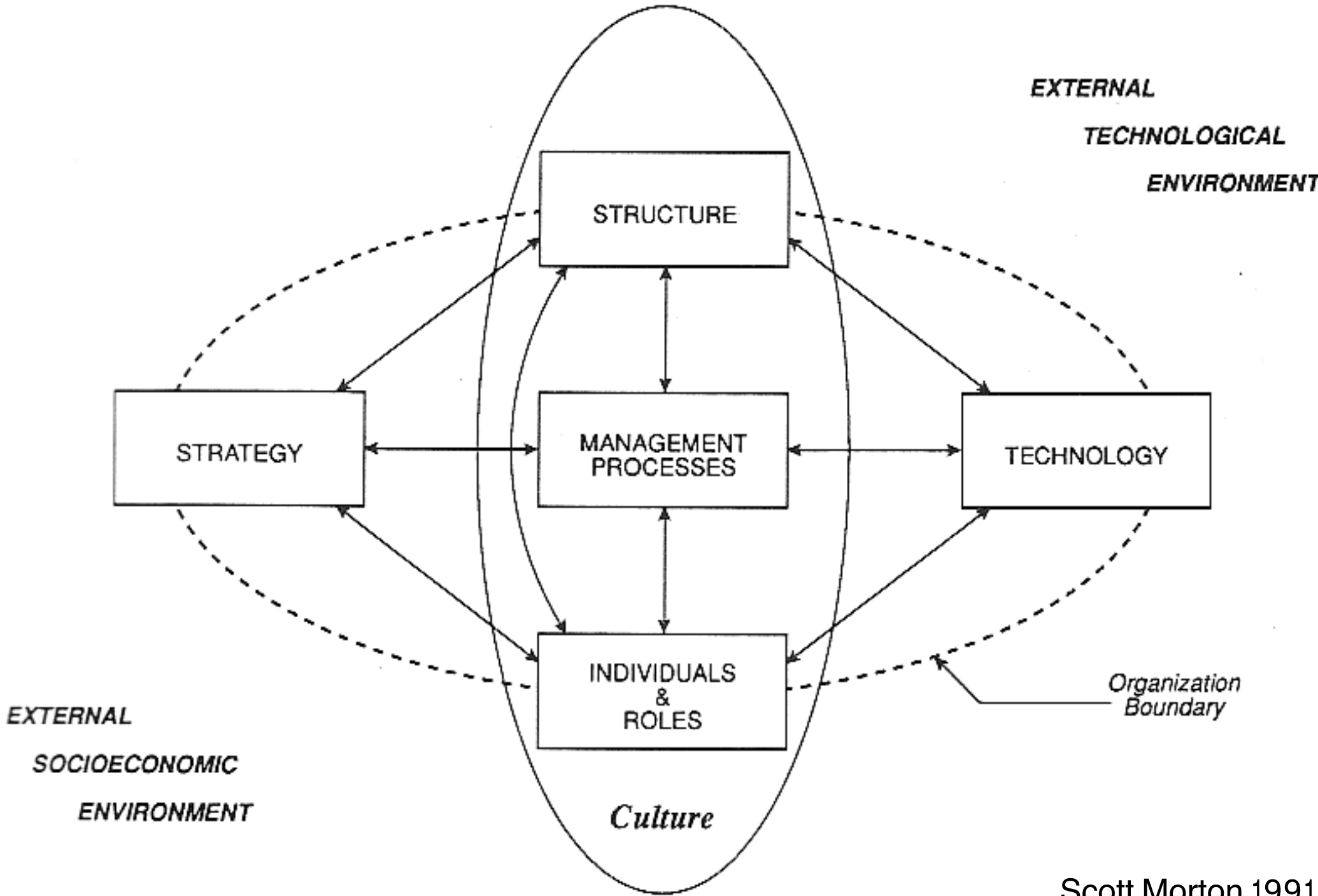
Customer Journey Mapping

Retail customer journey



Customer Persona: segments or personas to provide targeted insights. **Stages** : broken down into distinct stages, such as awareness, consideration, purchase, post-purchase. **Touchpoints:** points of interaction between the customer and business **Actions:** actions taken by the customer at touchpoints **Thoughts and Emotions:** The customer's thoughts and feelings at each stage. **Pain Points:** Areas where the customer experiences difficulties. **Opportunities** : areas for improvement based on the customer's experience. **Channels:** used by the customer during the journey.

MIT90's Model



Scott Morton 1991

Adapt or Die

Step 4—Adaptability:

- Time required for many tasks greatly decreased.
- Many tasks completely eliminated.
- Inventory and working capital dramatically reduced.
- New alliances, products, and revenue opportunities possible.

Step 3—Collaboration:

- Sharing of customer demand data with members.
- Targeted replenishment of supplies.
- Responsibility transferred to supply replenishment vendors.
- Ability to reallocate inventory to fulfill maximum number of orders.

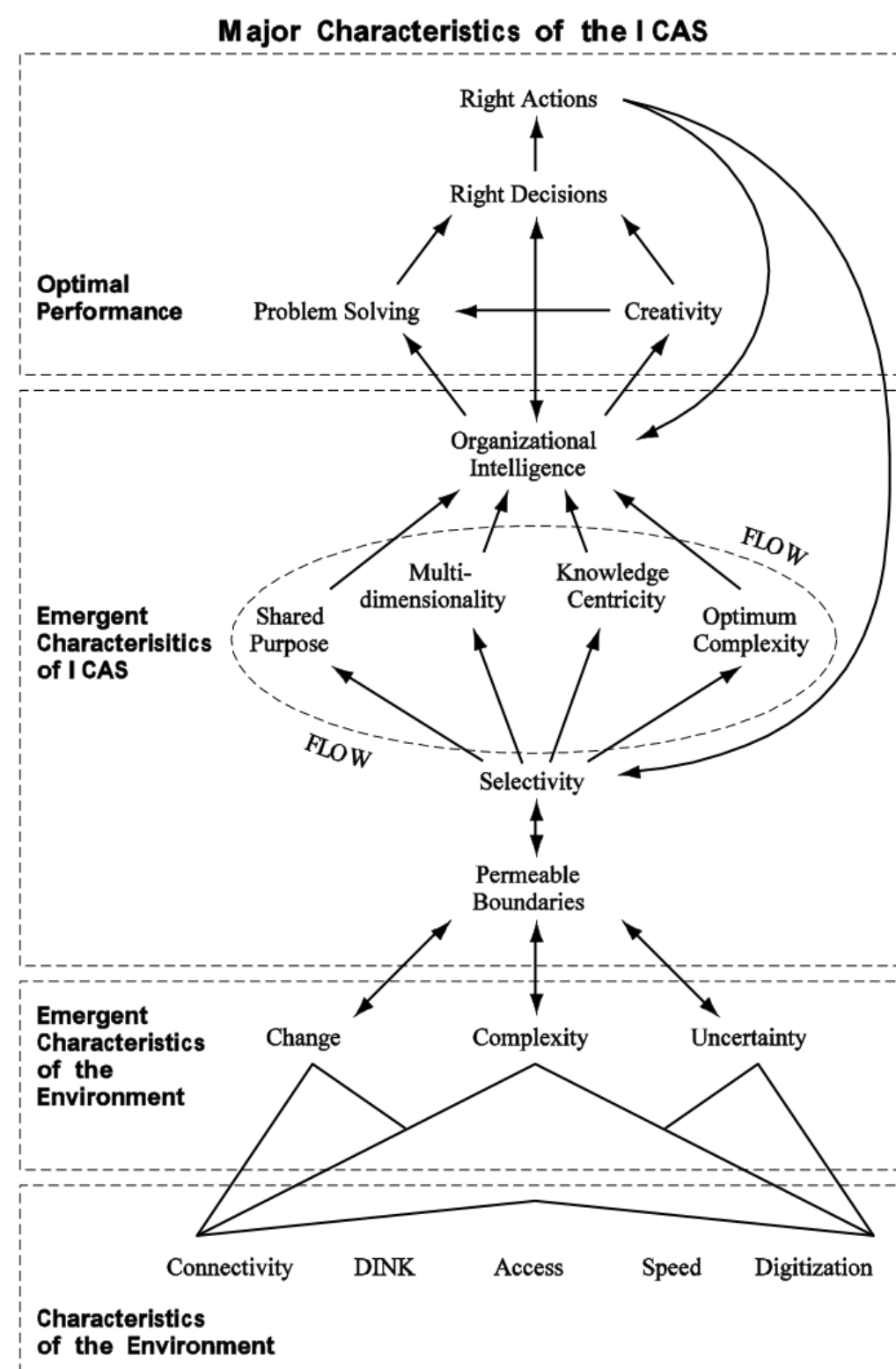
Step 2—Community:

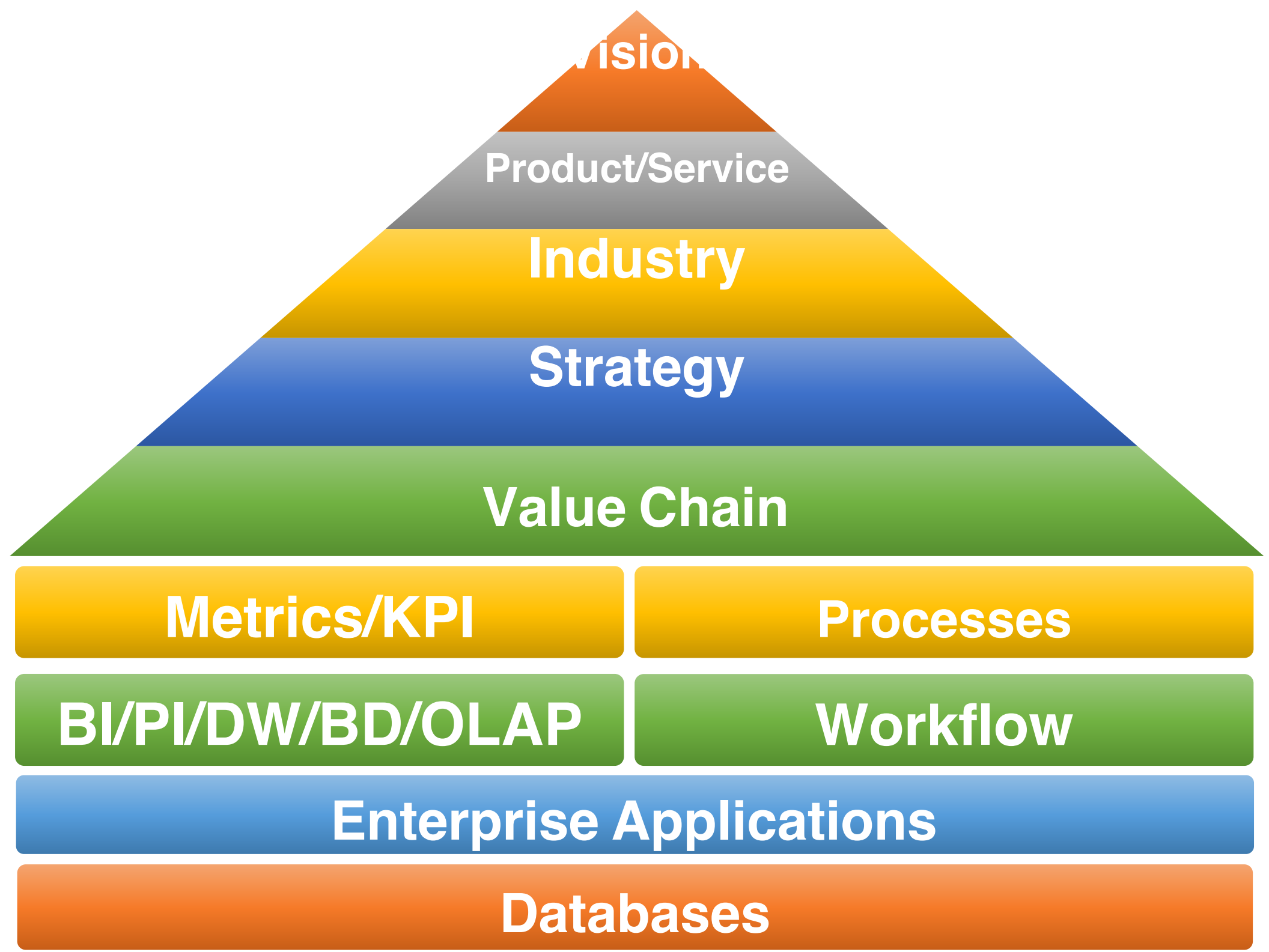
- Moving day-to-day transactions to the portal.
- Establishment of minimum and maximum control thresholds.
- Reduction of inventory.
- Time savings via mechanizing transactions.

Step 1—Visibility:

- Sharing of information with partner companies.
- Many routine business processes with partners standardized.
- Information posted for common view on portal.
- Greater insight into business process and data accuracy problems.

Organisational Survival in the New World: The Intelligent Complex Adaptive system





Vision

Product/Service

Industry

Strategy

Value Chain

Metrics/KPI

Processes

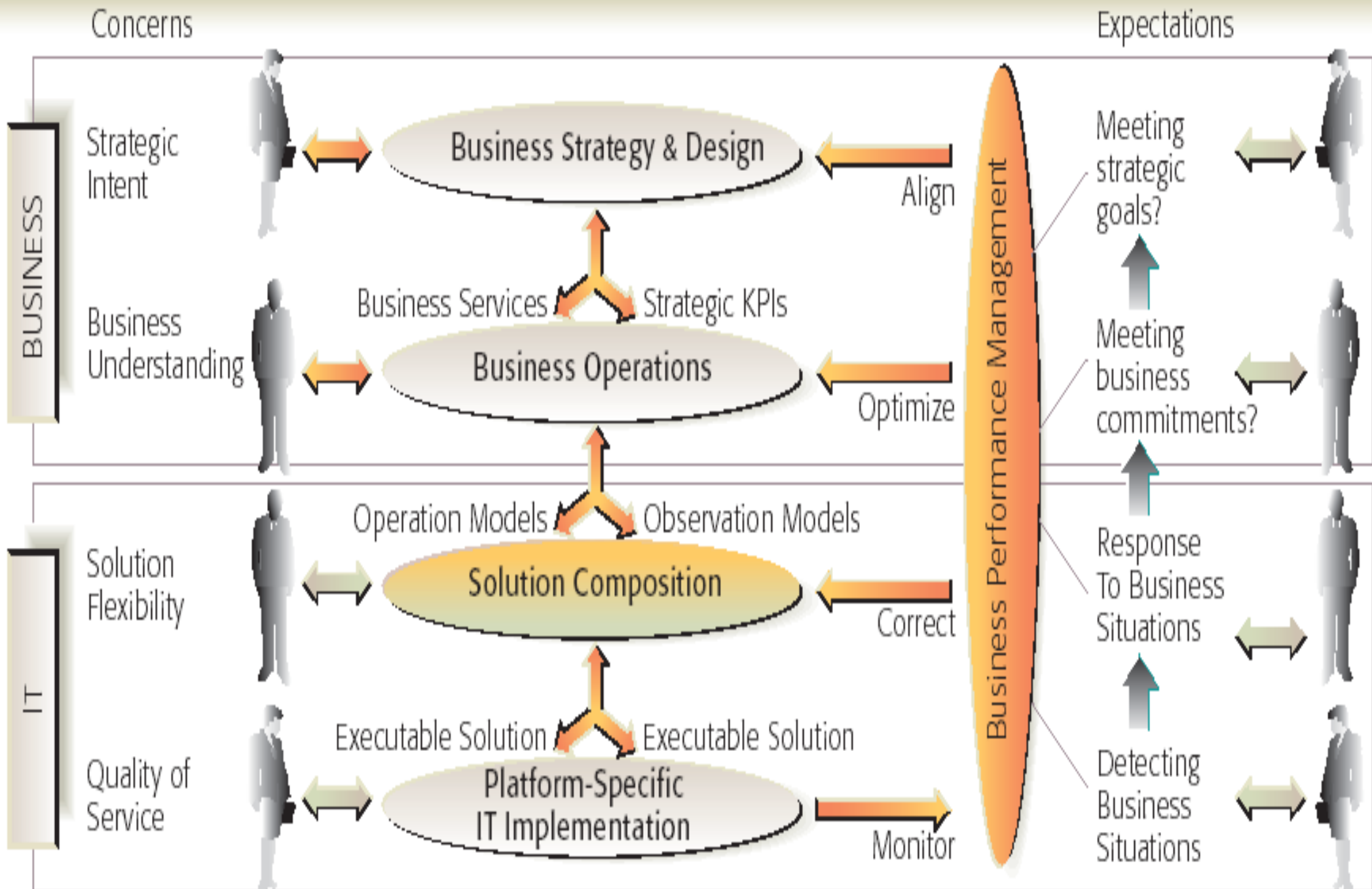
BI/PI/DW/BD/OLAP

Workflow

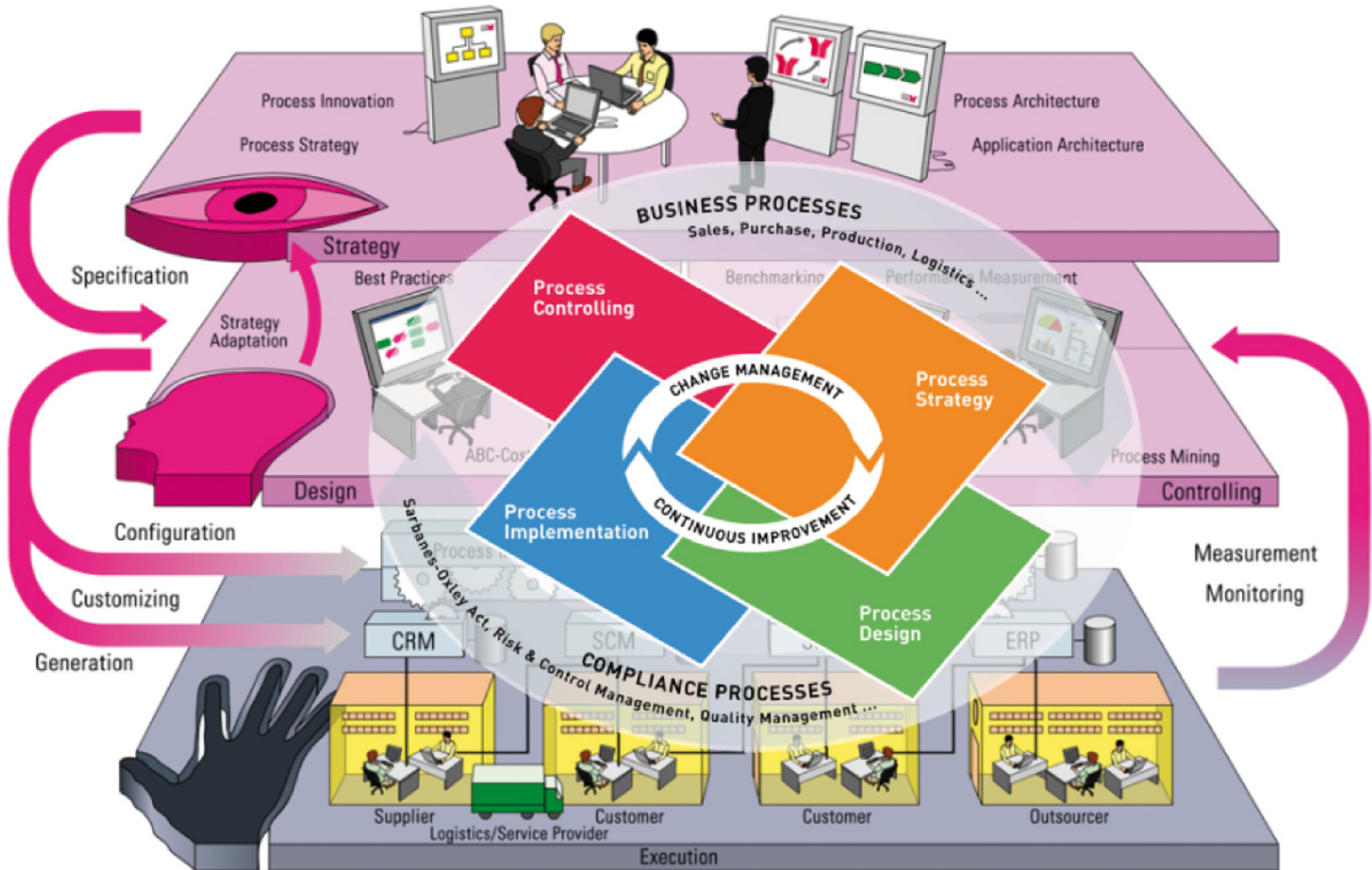
Enterprise Applications

Databases

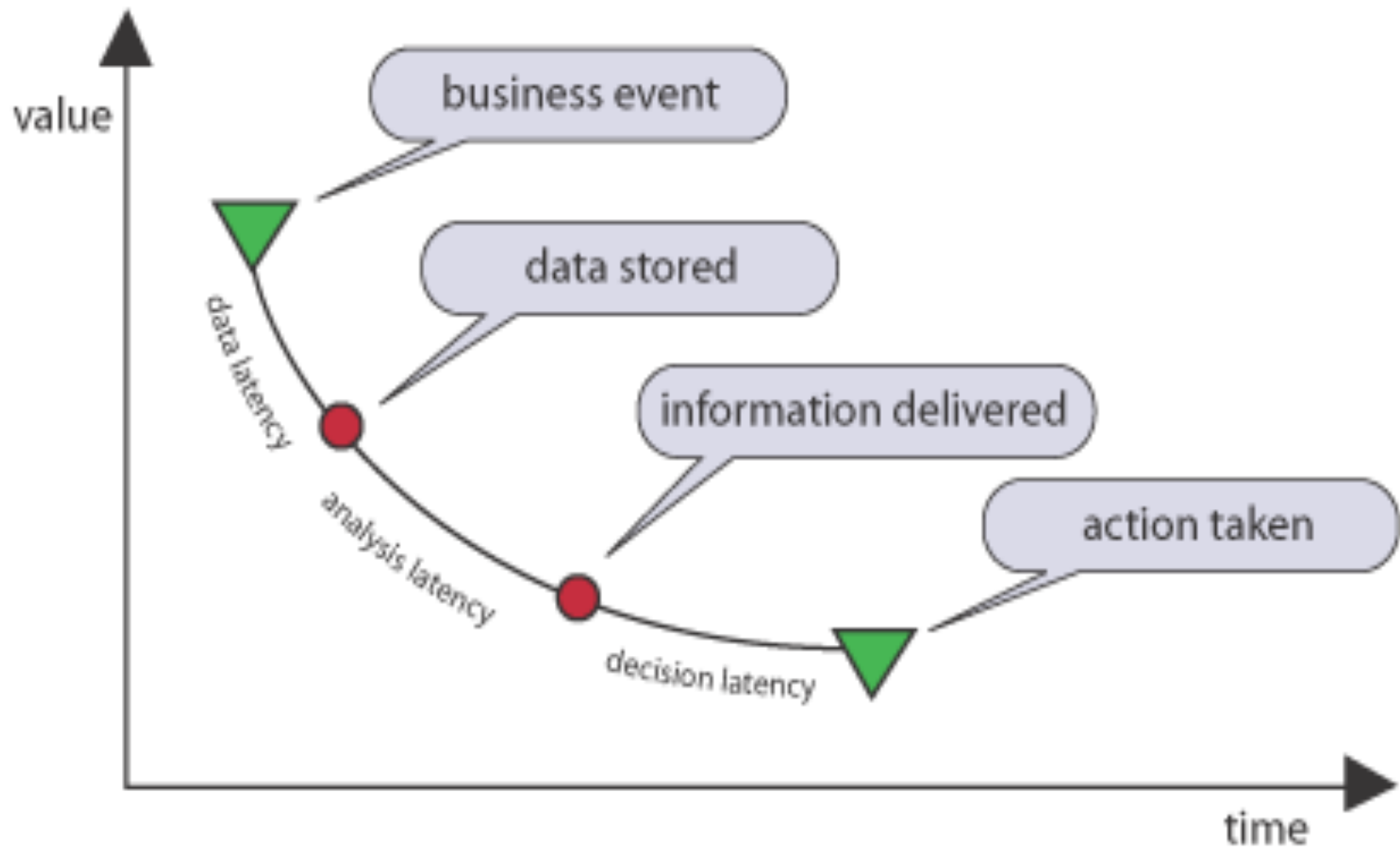
Adaptive Enterprises by Kumaran et al



BPM by Professor Scheer of ARIS

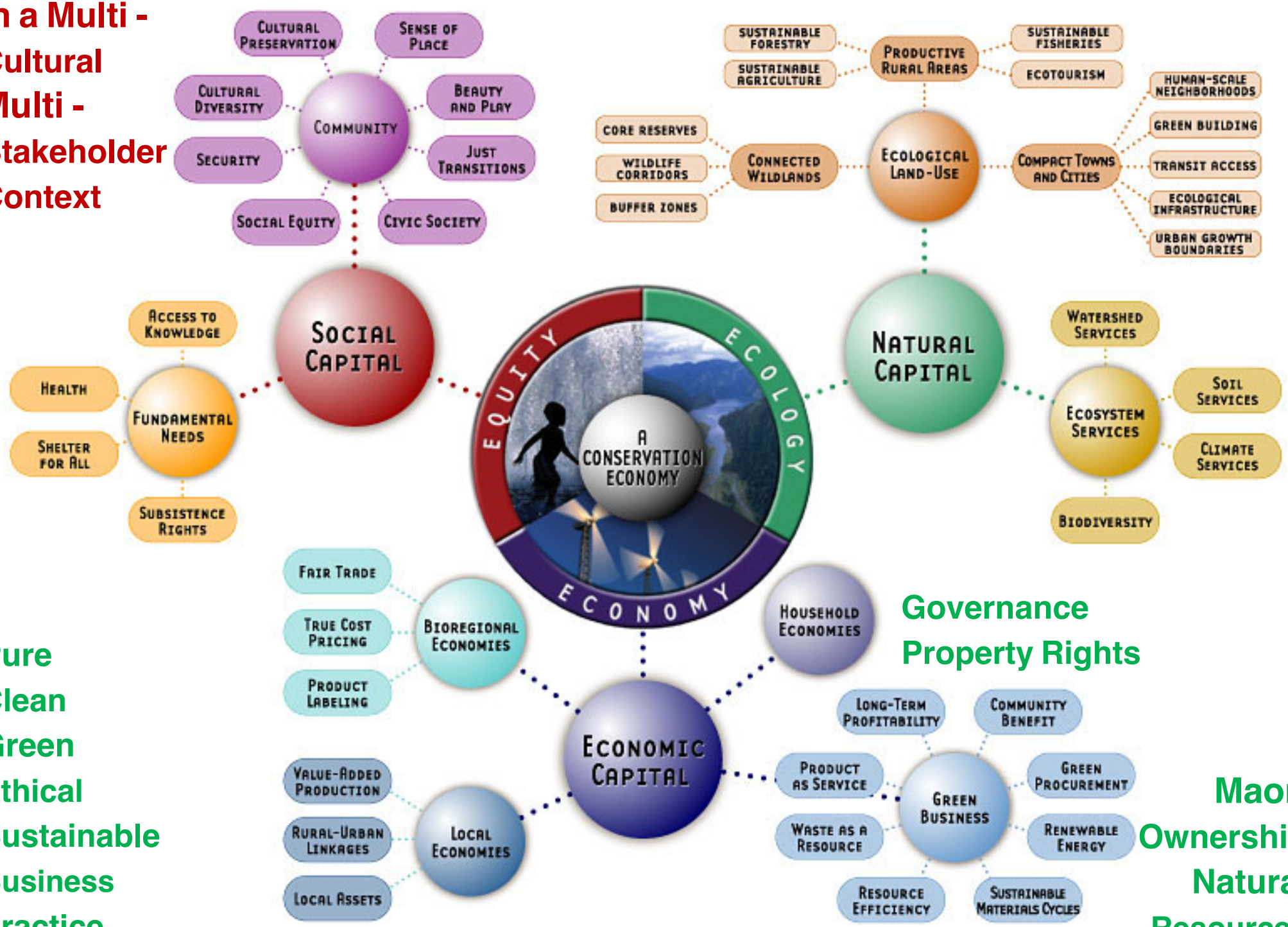


Latency



Role of Business
in a Multi -
Cultural
Multi -
Stakeholder
Context

Pure
Clean
Green
Ethical
Sustainable
Business
Practice



PESTLE

POLITICS

- Bureaucracy
- Corruption
- Freedom of the Press
- Government Type
- Government Stability
- Social/Employment Legislation
- Trade Restrictions

ECONOMICS

- Interest rates
- Taxes
- Exchange rate
- Economic growth – GDP, FDI
- Employment rate
- Likely Economic Change
- Big Mac Index

SOCIAL

- Health
- Population Growth Rate
- Age Distribution
- Career Attitudes
- Educational Infrastructure
- Social Mobility
- Employment Patterns
- Cultural Taboos
- Core Ethics

TECHNOLOGICAL

- Degree of Automation
- Emerging Technologies
- Rate of Technological Change
- R&D Activity

LEGAL

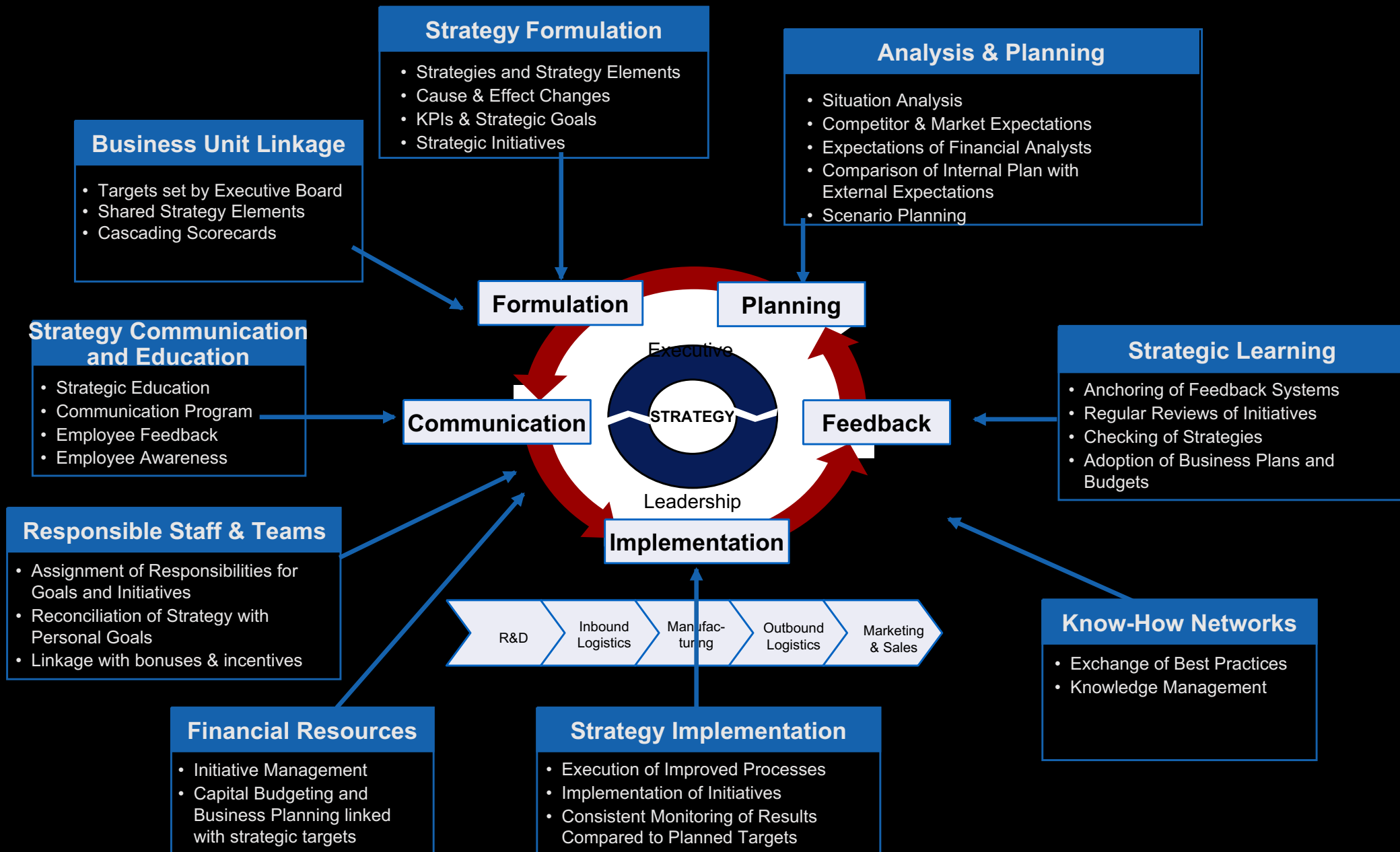
- Antitrust Law
- Consumer Law
- Discrimination Law
- Employment Law
- Health and Safety Laws

ENVIRONMENTAL

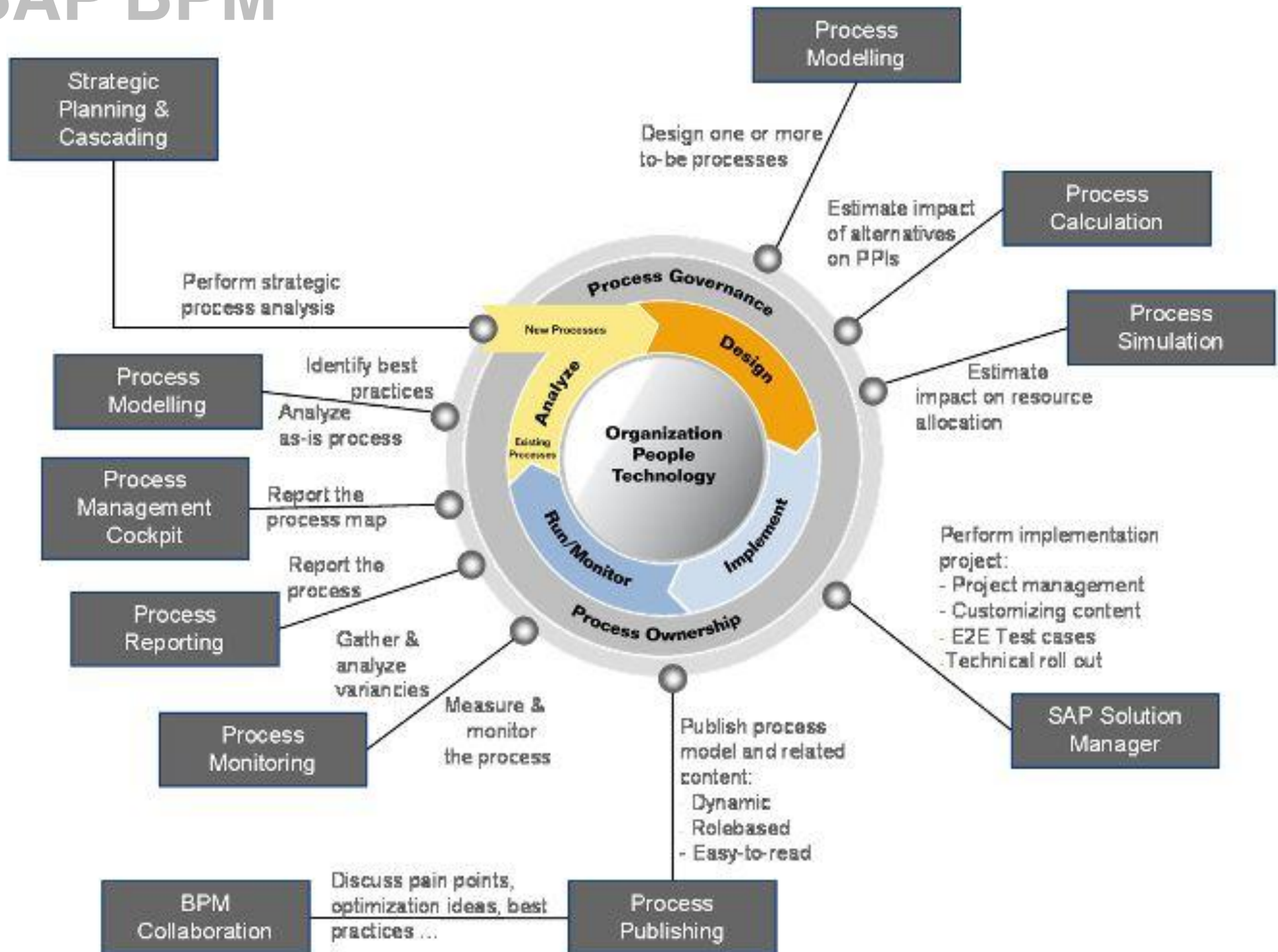
- Weather
- Climate
- Geography
- Disaster Quotient
- Infrastructure
- Legal



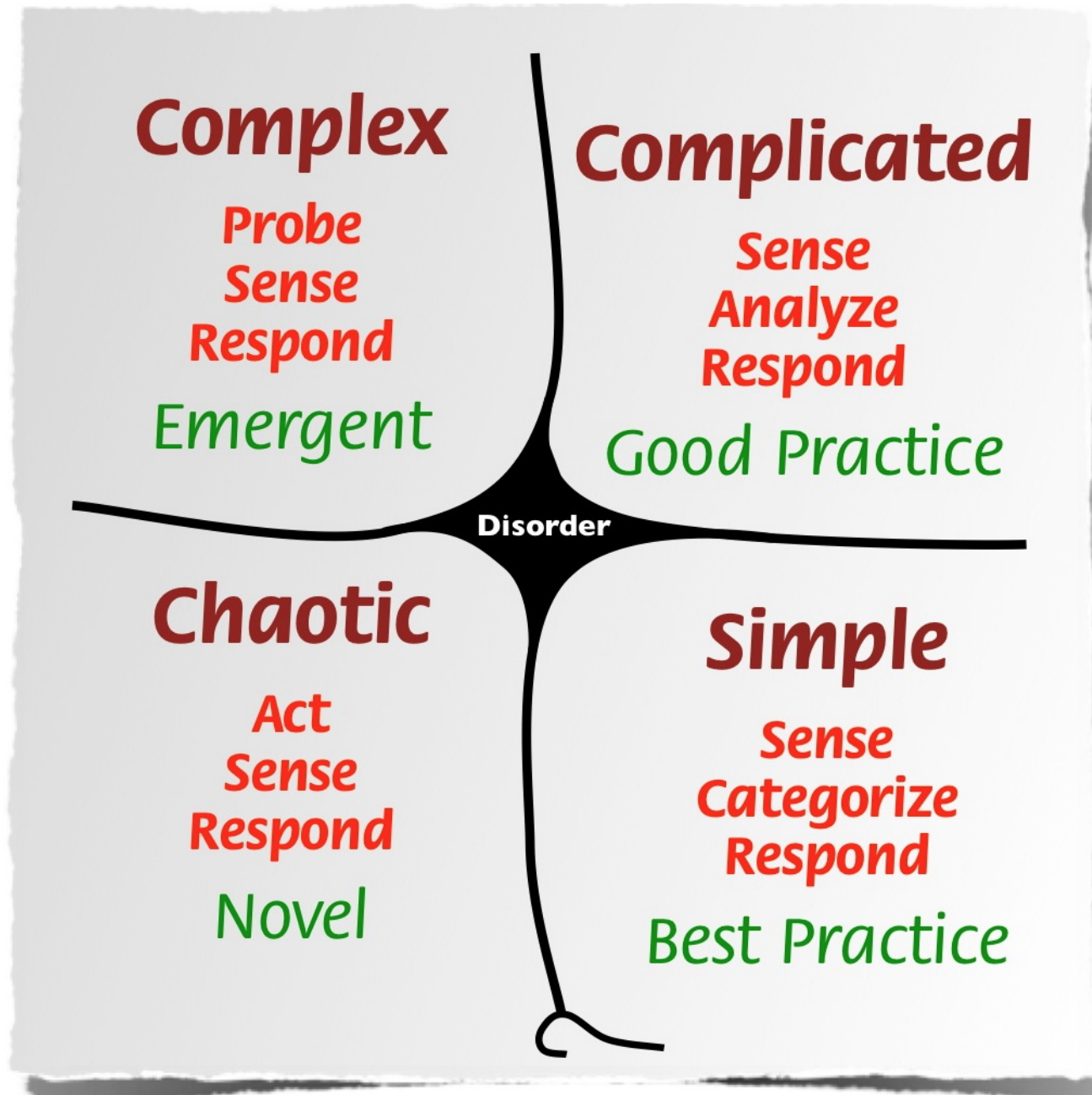
Strategic Enterprise Management by SAP



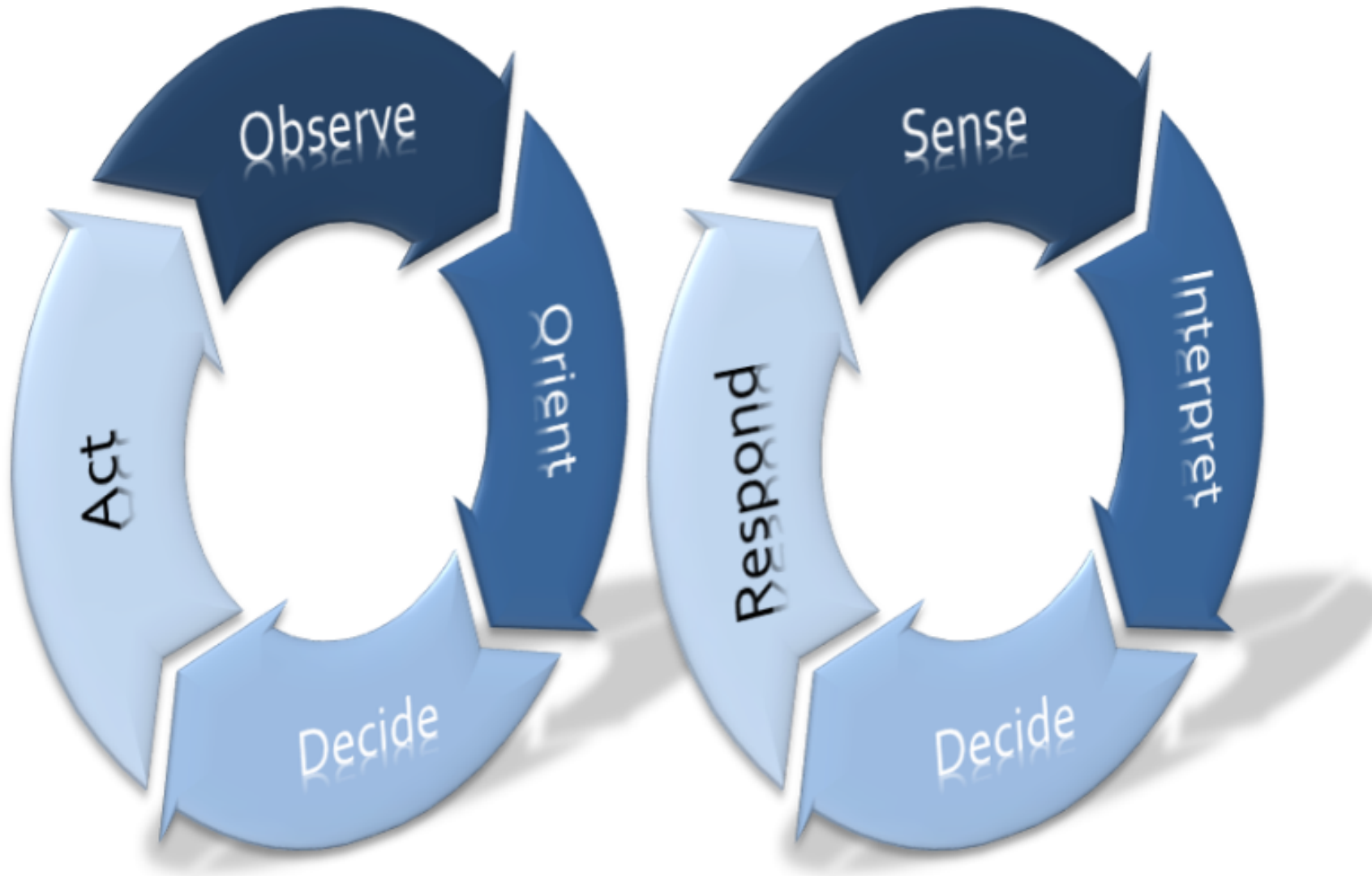
SAP BPM



Cynefin by Snowden



OODA and SIDR Loops



Sense and Respond

MAKE-AND-SELL

Assumption:

Predictable change

Goal:

Become an

Efficient Enterprise



SENSE-AND-RESPOND

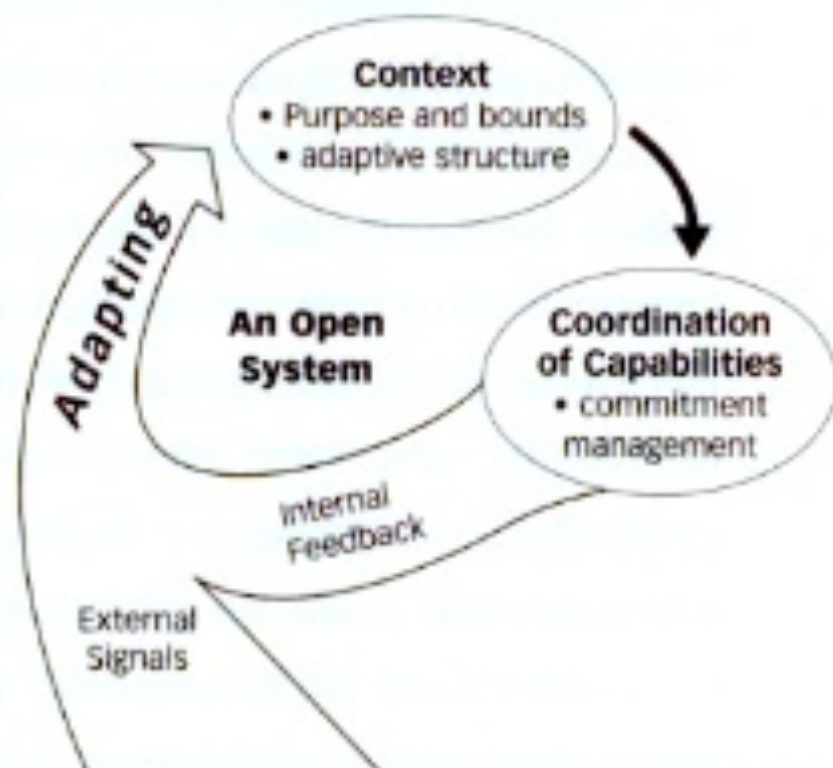
Assumption:

Unpredictable change

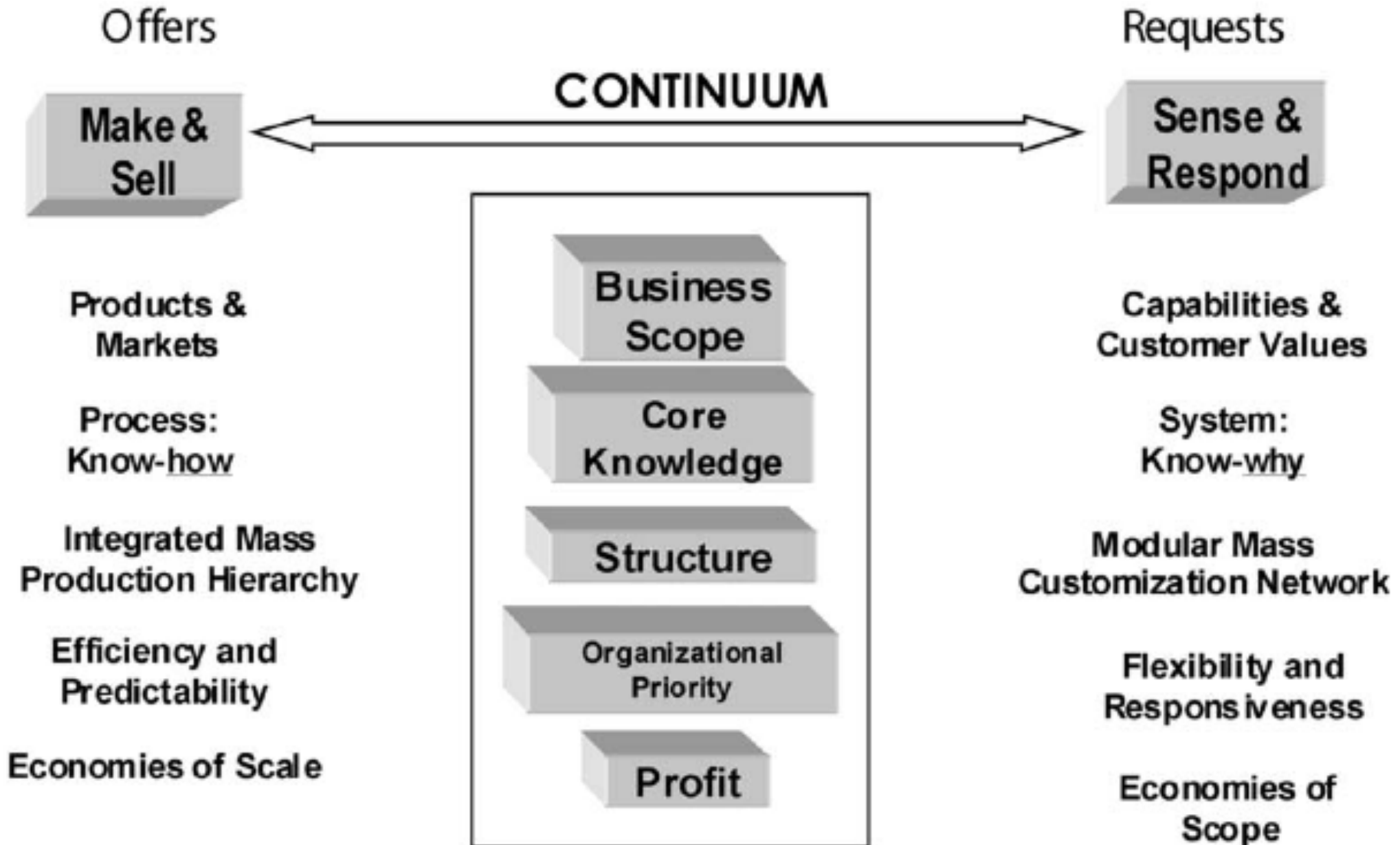
Goal:

Become an

Adaptive Enterprise

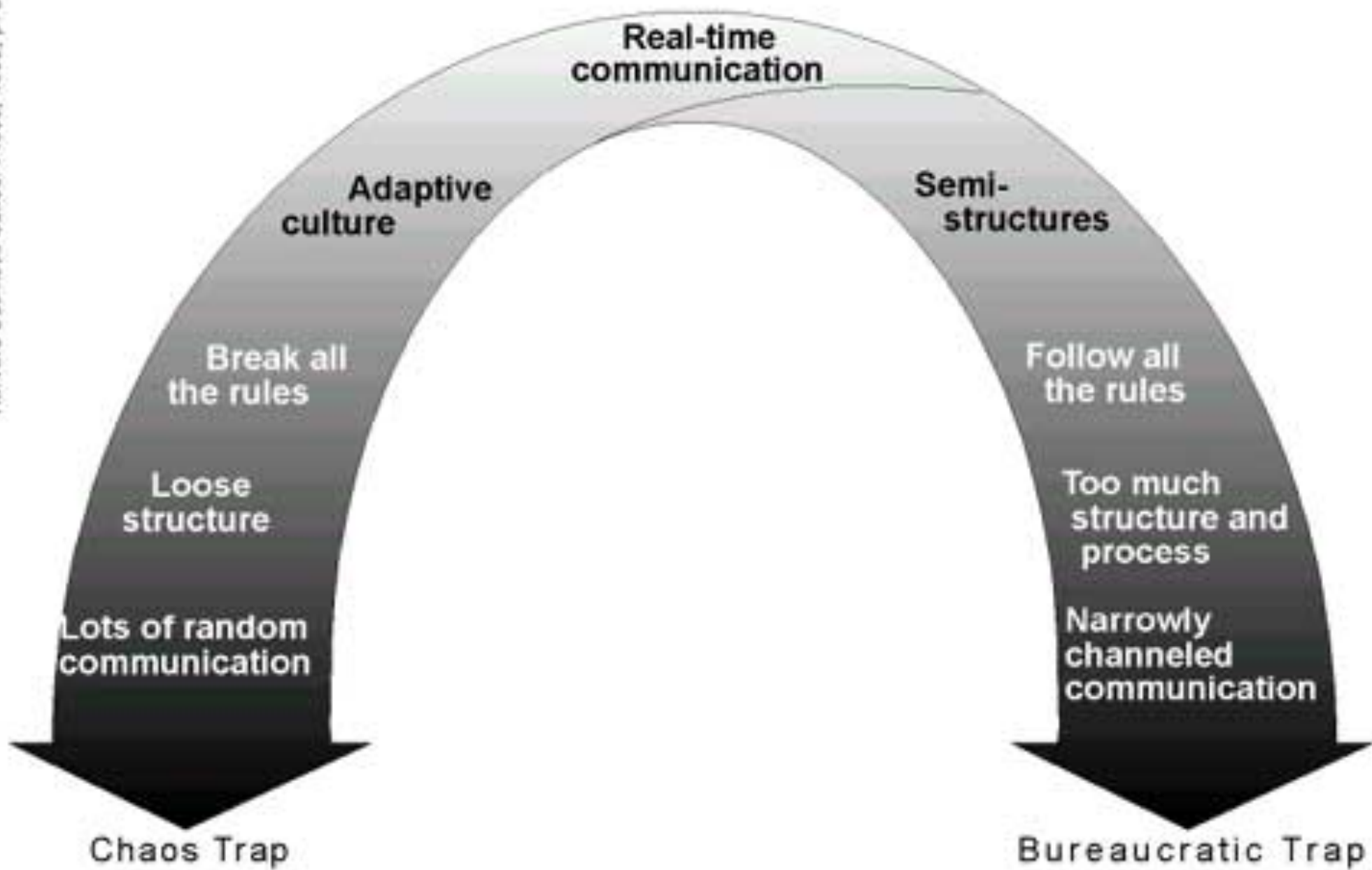


Make to Sell vs Sense and Respond



Competing on the Edge

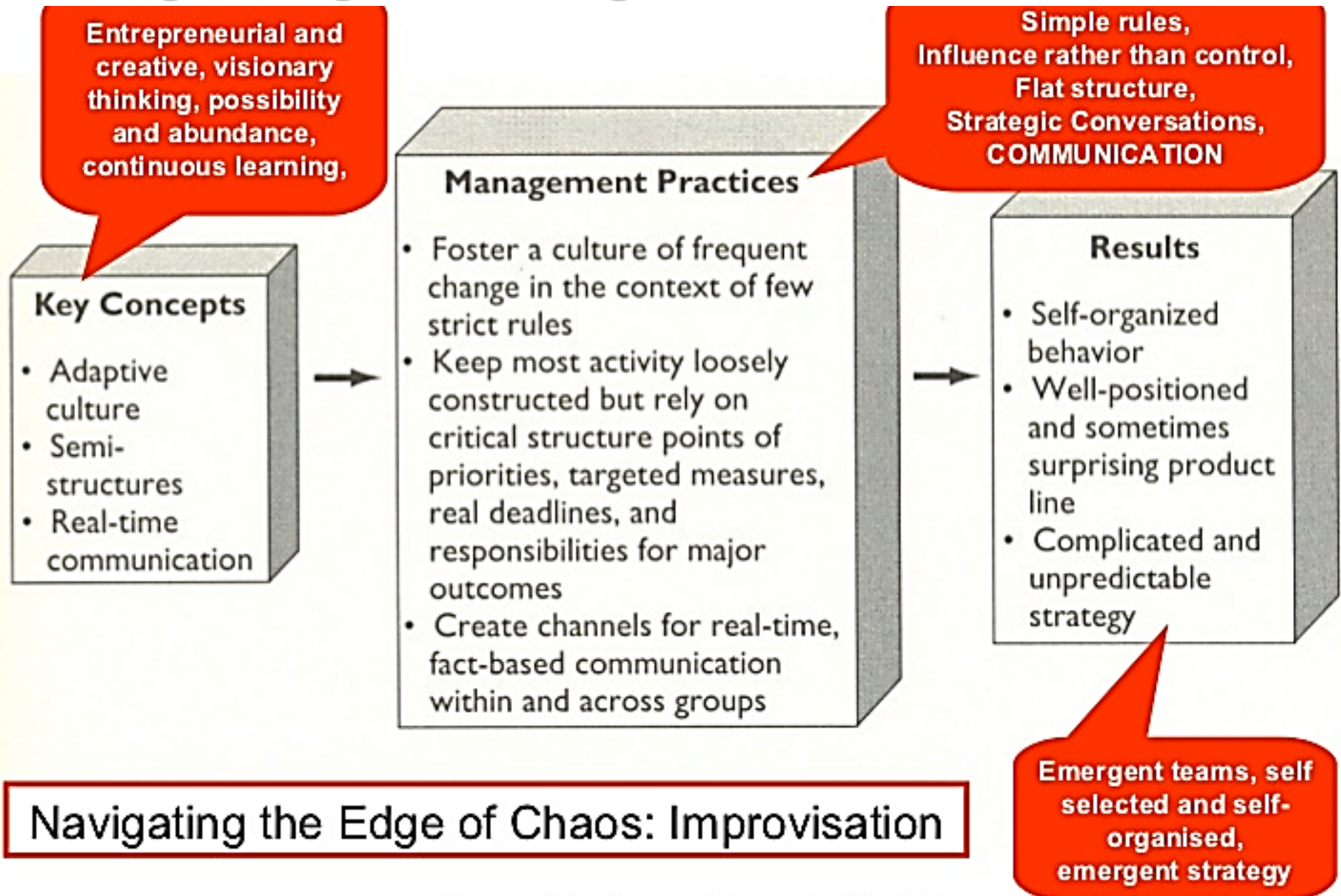
Source: Brown S. L./Eisenhardt K. M. - Competing on the Edge: Strategy as Structured Chaos, Boston: Harvard Business School Press, 1998, p. 30



Dilemma: Adaptively innovate and consistently execute

Improvisational Edge

Navigating the Edge of Chaos



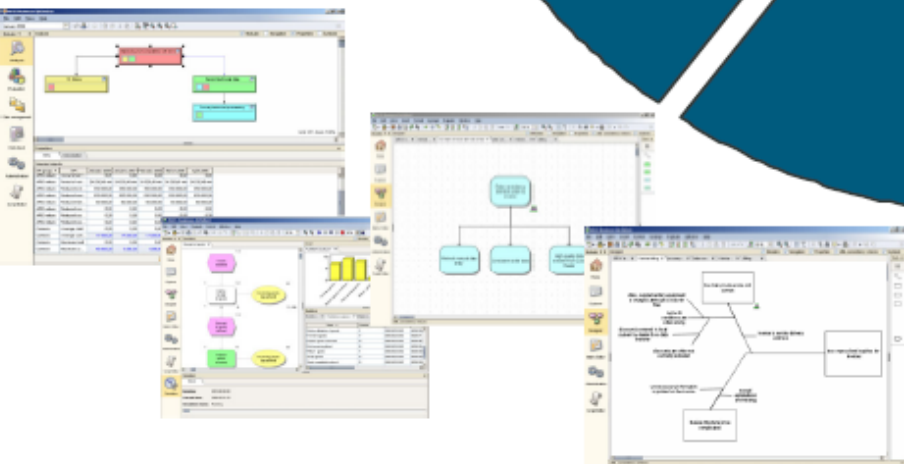
(Source: After Brown and Eisenhardt: 1998: 47)

ARIS Six Sigma

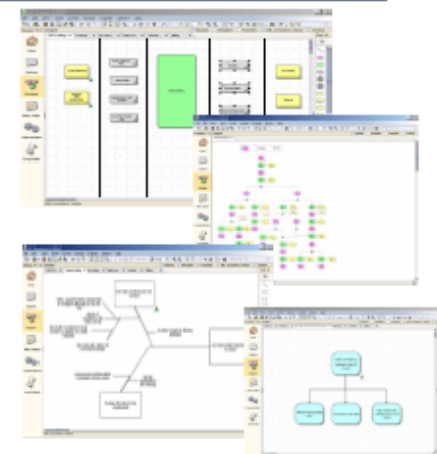
- Control the process with statistical methods
- Process capability, ...



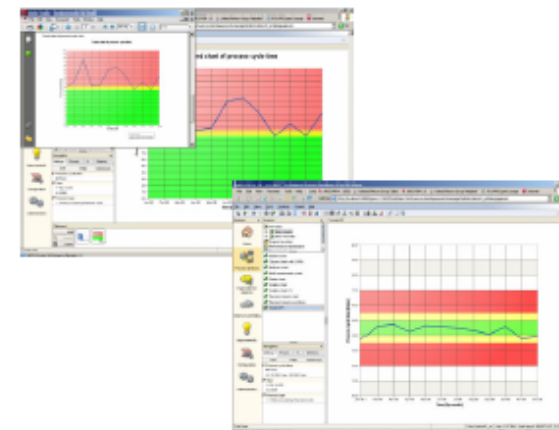
- Improve and optimize the process
- Process Variants, ...



- Define project (Problem, process improvement goals, team, time-plan, milestones)
- Document Processes
- SIPOC, Process Flows, CT-x table, Fishbone, ...



- Measure how well process really fulfills existing customers
- Processes, ...



- Analyze process data & main causes of defects
- Determine possible areas of improvement / gaps
- Fishbone, Process Flows, CT-x table, ...

Business Performance Management

Business Performance

STRATEGIC









TACTICAL

Sales

Cash Sales

Prozessanzahl: 115.00 / 115.00 %

Auftragsvolumen: 115.000.00 / 115.00 %

Standard Orders

Auftragsvolumen: 2.740.316.40 / 94.03 %

Prozesslaufzeit: 12.25 Tag(e) / 49.99 %





Performance Measurement + Process Discovery

OPERATIONAL

Status	Call-ID	Priority	Function
Open	30000070244	high	Open
Open	93352373	very high	Open
Open	93442423	normal	Open
Open	93562435	normal	Open
Open	93622341	normal	Open

Process view

Calls currently exceeding reaction time

Priority	Count	Average time complete
very high	1	20min. 33.40s
high	4	20min. 40s
normal	5	20min. 45.40s
low	8	20min. 54.40s



Business Events
Exceptions & Alerts
Real-time Process Instances

Management

Process Owner

Process Staff

How could you organize this?

The whole framework-driven approach is really trying to think about, “How could you organize this?” Every framework—all the way down to the simple two-by-two matrices we use day in, day out—are an attempt to frame the problem around some nifty set of three, four, or five balls or boxes or triangles or whatever you need to create a simple representation of a complex problem. McKinsey was masterful at that. I’ve really tried to adapt that for my work.

I find that, although frameworks work great internally at McKinsey, when you go outside McKinsey you have to be careful about their use. Many people will see a framework and automatically start getting defensive. We heard it a lot at McKinsey: “Oh, you’re taking an approach that you used on somebody else’s problems and trying to apply it to me. My problems are different.” We knew that wasn’t the case; we were just trying to get the thoughts started, giving ourselves a systematic checklist for what the key issues are and how to present those key issues. You have to be careful about introducing frameworks because they can carry a fairly negative connotation, especially if they are overused. Instead of reusing an old framework, use the concepts from the framework to generate new ideas that help solve the problem at hand.

Hypothesis

Having reduced the problem to its essential components through the use of appropriate frameworks, you are ready to embark on the next step in the process of framing it: forming a hypothesis as to its likely solution. McKinsey believes, and the experiences of McKinsey alumni demonstrate, that using an initial hypothesis to guide your research and analysis will increase both the efficiency and effectiveness of your decision making.

Solve the problem at the first meeting!

- Generate initial hypothesis based on limited facts without doing too much research. Maybe news articles, annual reports, etc.
- Identify which analysis you have to conduct and what questions you need to ask to PROVE or DISPROVE your hypothesis.
- Issue/Logic trees bridges the gap between frameworks and hypotheses!
- Brainstorm with your team to generate initial hypothesis
- All team members come to the brainstorming meeting well prepared with the known facts. Maybe even develop individual level hypothesis.

White Room Brainstorming

- Everyone should be able to express their ideas
- Leave preconceptions and prejudices at the door
- There are no bad ideas or dumb questions
- Be prepared to 'kill your babies'/get your ideas shot down!
- Don't allow brainstorming to drag on forever (30-60 minutes)
- Get it down on paper/system.
- The problem is not always the REAL problem! Dig deeper, ask questions, get facts, be skeptical.

Lessons Learned

- Hypothesis based decision making will be more effective.
- Avoids paralysis through too much analysis.
- Back of the envelope analysis.
- Helps focusing and prioritizing.
- Don't ignore instinct, intuition, and gut feeling.
- An initial hypothesis will save you time.
- Quick and Dirty Test QDT – what are your assumptions for the hypothesis to be true? Are these assumptions true?
- Weed out invalid hypotheses using QDT.

AS-IS Modelling

The Process Lifecycle

Process identification

Process modelling (as-is)

Process analysis

Process improvement (to-be)

Process implementation

Process execution (ES enabled)

Process monitoring/controlling

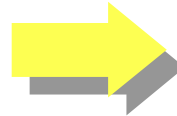
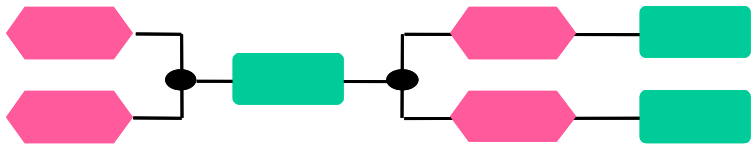


Business Process Modelling

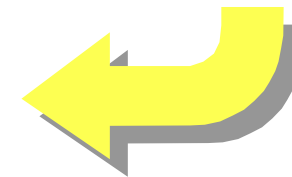
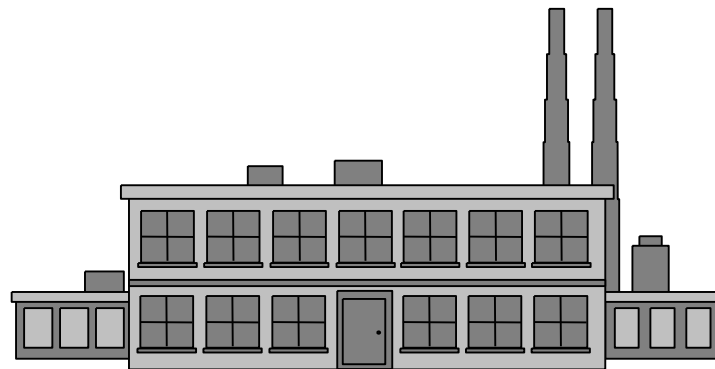
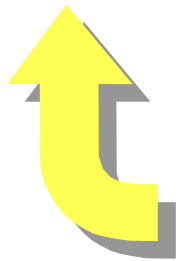
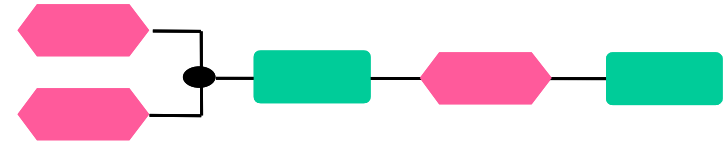
- ***Business process modeling*** (or *mapping*) - the activity of creating a detailed flow chart or process map of a work process showing its inputs, tasks, and activities, in a structured sequence.
- ***Business process model*** - a graphic description of a process, showing the sequence of process tasks, which is developed for a specific: (a) **AS-IS** process model (b) **TO-BE** process model.
- You cannot **TRANSFORM** what you have not **MODELLED!**

Objectives of Process Modelling

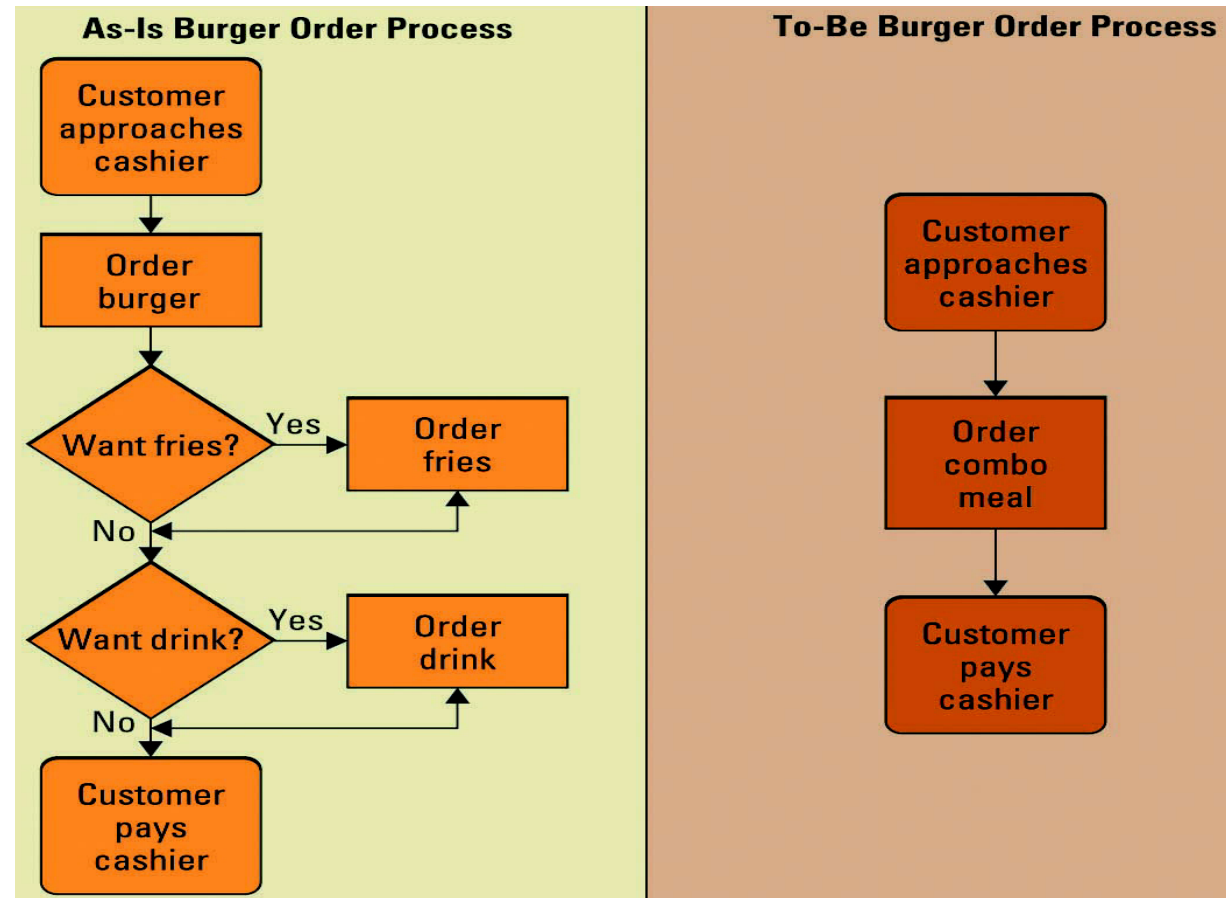
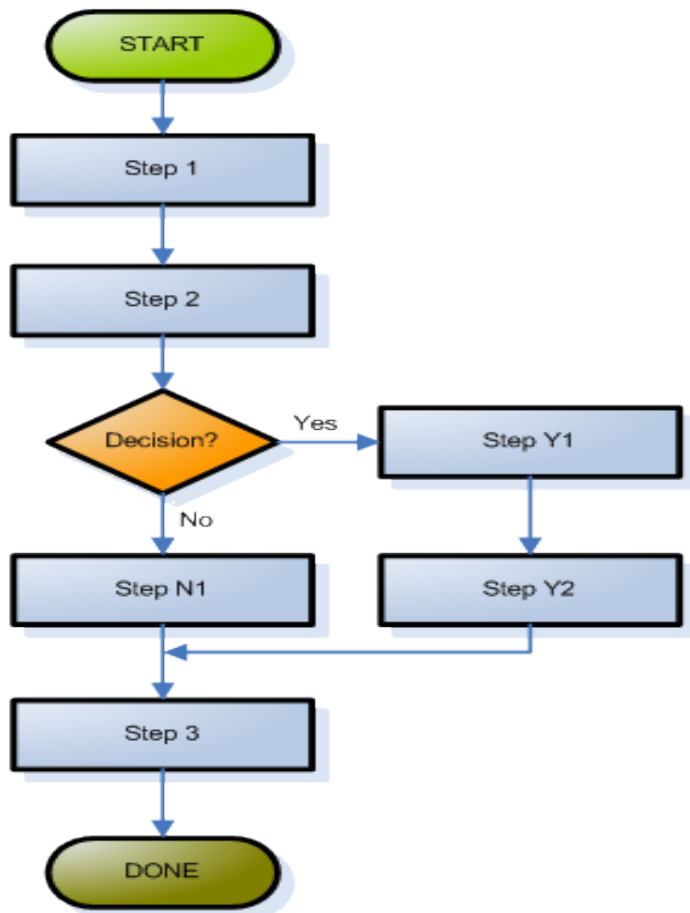
**Descriptive modeling
of the real world (as-is)**



**Prescriptive modeling
of the real world (to-be)**

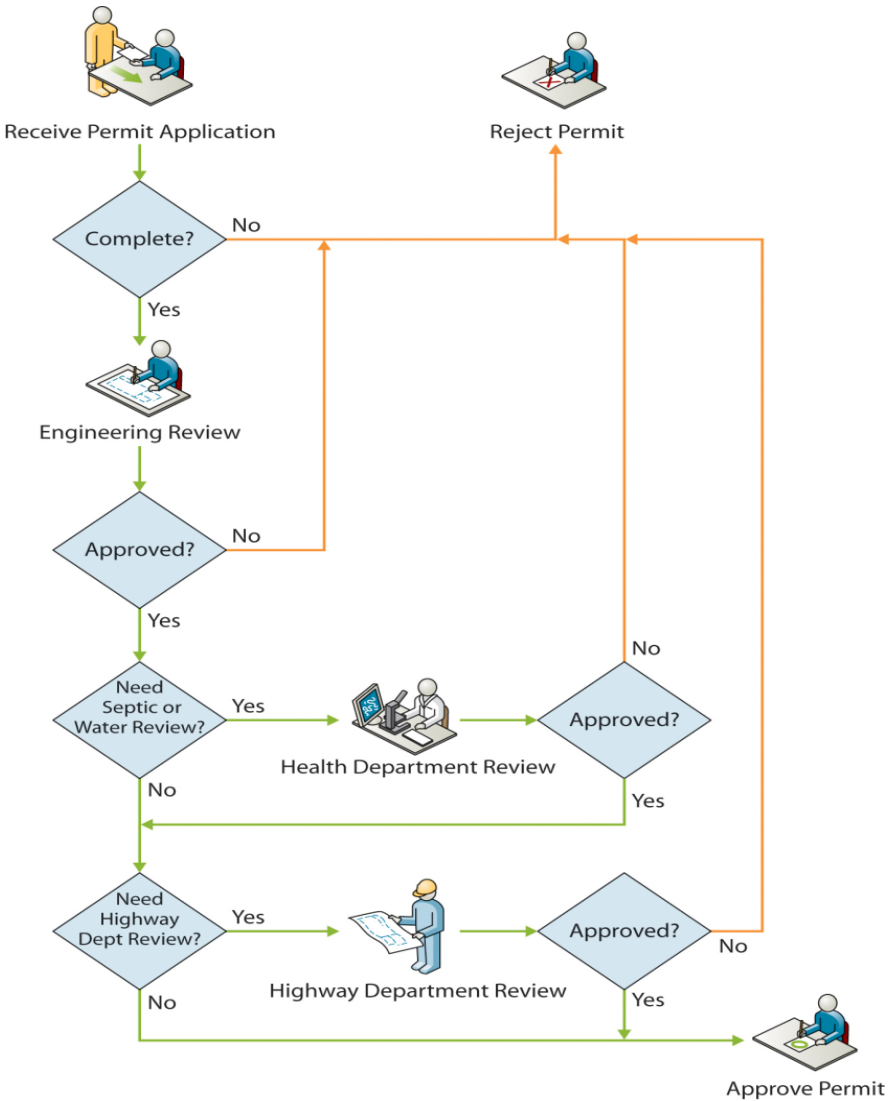
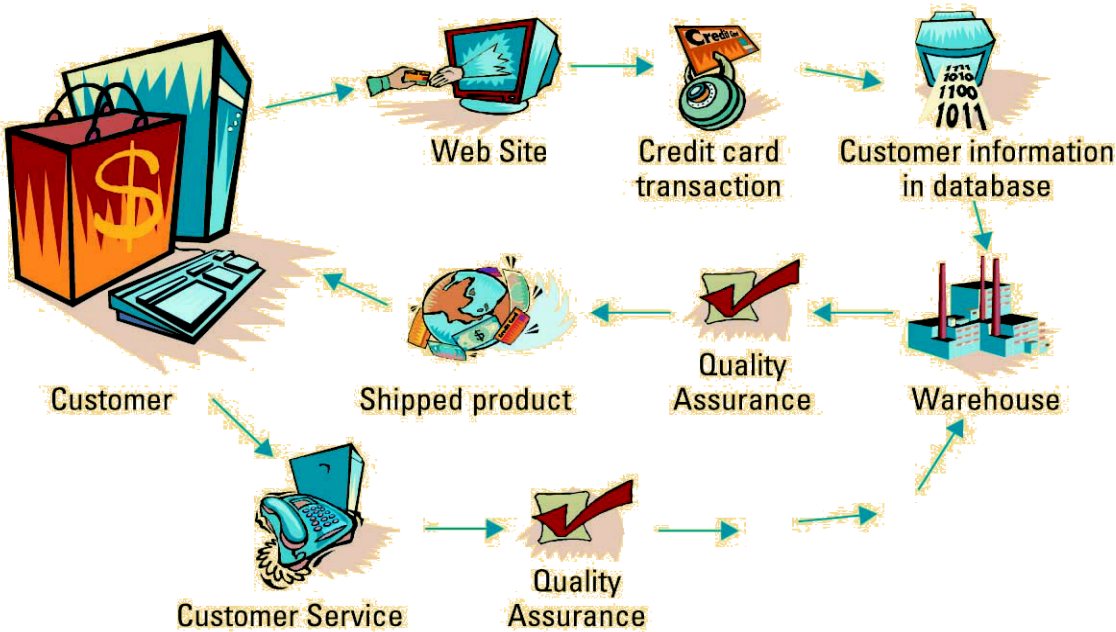


Business Process Modelling

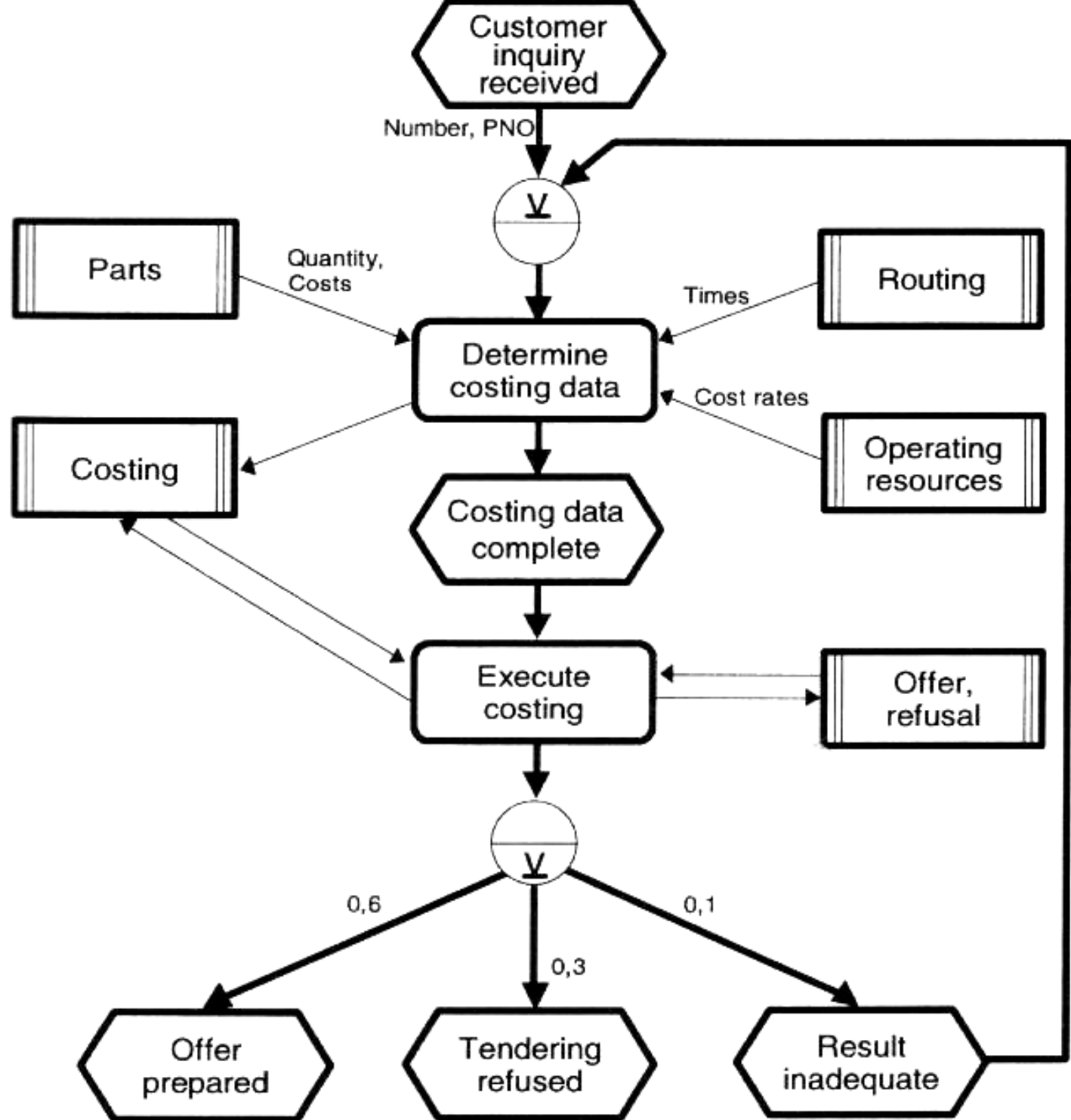


Business Process Modelling

E-Business Process Model



Order Processing with Data Flows



Legend:

→ Control flow

→ Data flow

Reasons for Modelling	Aspects to Consider
Business Planning	Concentrate on business objectives, customer needs and metrics. Use <i>Value Chain Diagrams</i> , <i>Balanced Scorecard Models</i> . Look at high-level business functional breakdown.
Business Restructuring	Concentrate on the organisations that are carrying out tasks and the hand-offs between them. Look at the value added by each Function. Use <i>eEPCs as row display</i> to provide organisational swim-lane view.
Baseline the Business	Usually impractical in all but the simplest and static businesses. It takes too much time and the world moves on in the meantime. Concentrate instead on key processes, processes that have to change or where you already know there are problems. "Don't model the Universe".
Operational Process Design	Concentrate on getting the flow of the process correct. Use <i>eEPCs</i> and <i>FADs</i> . Identify the key decisions being made. Look for failure paths as well as the normal process flow. Identify inputs and outputs for all tasks. Identify key documents and sources of information.
Systems Development	Requires very detailed logic flow to be modelled. Exception handling is very important. Detailed data models, data flow and systems interfaces must be modelled.

Modelling Viewpoint	Approach
Follow a data object	Possibly the easiest approach to take. Select a key data item (e.g. a customer order) and follow it through the process. See what actions are performed on it, who handles it and where it ends up. This is also useful for testing other model viewpoints.
Model the business	Modelling what the whole business does is one of the hardest approaches. It can normally only be done at high levels of abstraction and it is often difficult to identify the triggers and outcomes.
Model a business function	The most common approach is to model a particular business function (e.g. order-handling, fault-reporting, etc.). This will normally involve many different organisational units. Modelling organisational hand-offs will be essential. Can lead to very 'company oriented' models that forget about the customer.

Modelling Viewpoint	Approach
Model a business process	The most useful approach (but not often done) is to model the end-to-end processes that a business function performs. Particularly valuable when done from a customer perspective. Better than the business function approach as it helps ensure the whole process hangs together to deliver a good customer experience. Helps identify failure modes.
Model an organisation	Another common approach is to model what an organisation does. This may not necessarily be the most useful approach. Organisations change over time and the range of functions an organisation performs may have evolved historically.
Model an organisational unit	This model just focuses on what a single unit does. The model shows the interfaces with other units, but doesn't worry about how they accomplish their tasks. Provides a very focused model, but can over-simplify what is going on. It may also encourage an out-of-sight, out-of-mind approach, which doesn't spot gaps and failure points in the end-to-end process.

Modelling Viewpoint	Things to Consider
Business Unit	Business units (e.g. Sales, Manufacture, etc.) provide a pragmatic modelling abstraction. They are well understood and have major significance to the business. They do not change frequently, but when they do they have a significant impact on business processes which you have to reflect in your model.
Line Management Team	To be avoided at all costs. Have very little business significance, change frequently and have little impact on processes when they do.
Operational Centre	A very useful modelling abstraction. Normally have a very significant impact on process, change infrequently and fit within Business Units. Consider whether it is sufficient just to nominate the operational centre that does a task (e.g. Sales Office) or whether it is necessary to be more explicit about the roles within the centre (e.g. Sales Office Customer Service Advisor).

Modelling Viewpoint Management Layer

Roles

Things to Consider

Many business sectors have functional layers. For instance in Telecommunications we talk of Service Management, Network Management, etc. They can provide a useful level of abstraction, but can be problematic because often there is no clear definition of what they mean. Useful when modelling 'to-be' scenarios, but Business Units are better for 'as-is' models.

The lowest practical level of organisational unit. If used in process models you should make sure they are unique. The role "Customer Service Advisor" or "Planner" when attached to a Function doesn't convey very much as many operational centres will have such roles. You can model their parentage in an organisation chart, but better to make the names unique and meaningful (e.g. Sales Office – Customer Service Advisor).

Modelling Viewpoint	Things to Consider	
'as-is'	The process as it is now. But be careful to define what 'now' means. Does it mean as documented now or as operated now? If you know the documented process is wrong and a 'work-around' is being used, you have to decide which you are going to model. If changes are being made while you are capturing the process, are you going to include them or freeze the model at a certain point in time?	
'to-be'	How far in the future are you considering? Are you starting from scratch with a new business model or re-engineering what you already have? How radical can you be? What constraints have you got?	
Time-scale	Complex processes can sometimes complete within seconds, while simple processes can often last for weeks. What time-frame is important to your model?	
Delays	Are potential delays in the process important to you? Populating Function attributes with process times may not make delays explicit to people viewing the model. Consider explicitly modelling delays as additional Functions if you want to draw attention to them.	
Simulation	Simulation can be used to analyse process delays and optimise performance. A very powerful tool, but requires good quality models and good quality data. Models must conform to certain rules. Consider seeking advice from specialist simulation experts.	Davis,2001

Pitfalls of Process Modelling

- Lack of Strategic connections, Governance, Synergies, Qualified Modellers, Qualified Business Reps and User Buy-in
- Lack of Realism
- Chicken and Egg Problem
- Lack of Details/lost in it
- Lost in Translation
- Lost in a Drawing Tool
- Lack of Complimentary Methodologies

Pitfalls of Process Modelling

- L'Art pour l'Art
- Lost in Syntactical Correctness
- Focus on Models and not on Modeling
- Lack of Imagination
- Lost in Best Practice
- Design To-Be Models Solely Centered on New IT
- Modeling Success Is not Process Success
- Lost in Model Mtce

Designing the Analysis

Engagement Manager

- Analysis is a primary task of ALL consultants!
- Designing the Analysis also known as Work Planning usually the task of the Engagement Manager (EM).
- EM determines analyses to be done and who will do it.
- EM will discuss with team member the task, where to find data, and how the end product will look.
- Time is money – hence how to conduct rigorous analysis quickly?

Analysis Guidelines

- Find the key drivers
- What factors affect the problem the most and focus on those
- Drill down to the core
- Look at the big picture – don't miss the forest for the trees
- Don't boil the ocean! Don't be a perfectionist. Focus on analysis that absolutely NEEDS to be done to prove or disprove your hypothesis.
- IF you cannot identify initial hypotheses allow the solution to come to you as you progress.

Lessons Learned

- Let your hypothesis determine your analysis – balance intuition with data - figure where you want to go and then determine whether you are on the right track – quality over quantity – focused analysis
- Start with the end in mind – issue, sub-issue, hypothesis, analysis, data collection, end product.
- Get your analytical priorities straight – identify indispensable analysis – identify what NOT to do
- There is never enough data nor time HENCE take action earlier rather than later. Identify quick win/low hanging fruit analysis that follow the 20/80 Pareto rule.
- Forget about absolute precision – rough ballpark – don't get lost in the data
- Triangulate around tough problems – such as lack of data – find analogous situations

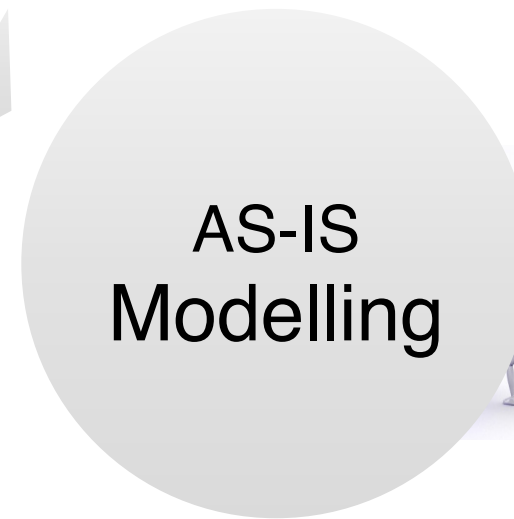
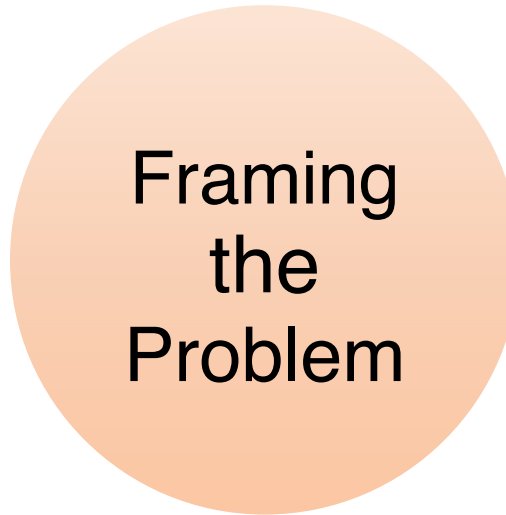
Implementation Guide

- Your initial hypothesis as to the answer
- The analyses that must be done to prove or disprove that hypothesis, in order of priority
- The data necessary to perform the analysis
- The likely sources of the data (e.g., Census data, focus groups, interviews)
- A *brief* description of the likely end product of each analysis
- The person responsible for each end product (you or a member of your team)
- The due date for each end product

Designing the Analysis Example

Table 2-1. Work Plan for Issue in Acme Widgets Issue Tree

Issue/Hypothesis	Analyses	Data Sources	End Product	Responsibility	Due Date
Can we implement the necessary changes to the production process? Yes					
Does the new process require special facilities? No	Technical Specifications	Articles, interviews	Chart	Tom	3-Jun
	List of facilities that meet new criteria	Facilities management, interviews	List	Tom	5-Jun
If it does require special facilities, can we acquire them? Yes	Map of “facilities gap”	Facilities management, thrum-mat line supervisors, interviews	Chart	Belinda	7-Jun
	Sources of required facilities/equipment	Operations, trade publications	List	Belinda	7-Jun
	Costs to fill gaps	Operations, contractors, interviews	Table	Belinda	10-Jun
	Effect on project rate of return	Finance department, prior analysis	Spreadsheet	Terry	12-Jun



Designing
the
Analysis

AS-IS
Modelling

Framing
the
Problem

Agenda